PROJECT USECASE SELECTION

UC01 – Claims Fraud, Waste and Abuse Risk Detector

This use case focuses on identifying potentially suspicious healthcare claims and provider billing patterns. The system analyzes claim-level and provider-level behavior to identify unusual activity, explain why a claim or provider appears high-risk, and prioritize cases for investigation. It is important for reducing financial leakage, but it requires careful handling of false positives because legitimate providers could be incorrectly flagged as fraudulent.

UC02 – Prior Authorization Triage and Policy Companion

This use case focuses on automating the initial triage of prior authorization requests. The system receives a healthcare request, extracts important clinical and administrative information, identifies the applicable insurance coverage policy, evaluates the request against the policy requirements, and recommends whether to approve it, pend it for nurse review, or request additional information. The reasoning behind the recommendation is also shown, making the process more transparent.

UC03 – Risk Adjustment and HCC Suspecting Assistant

This use case focuses on helping insurance organizations identify potential gaps in risk-adjustment documentation. The system analyzes synthetic member diagnoses, encounters, and claims, maps relevant diagnoses to Hierarchical Condition Categories, and identifies possible documentation or outreach opportunities. The key challenge is ensuring that the system only identifies conditions supported by available data and never invents a diagnosis.

UC04 – Star Ratings and Quality Gap-Closure Simulator

This use case focuses on improving healthcare quality performance for Medicare Advantage plans. The system identifies gaps in preventive care and chronic-condition management and prioritizes members who may benefit from intervention. It can also simulate how closing specific care gaps could potentially influence a plan's quality performance.

UC05 – Provider Network Adequacy and Access Intelligence

This use case focuses on understanding whether members have sufficient access to healthcare providers in their geographical areas. The system analyzes provider availability by location and specialty, identifies potential access gaps, and recommends areas where the insurance organization may need to add providers. It combines provider, geographic, specialty, and utilization information.

UC06 – Pharmacy Formulary Optimization and Adherence Assistant

This use case focuses on helping insurers manage their prescription-drug benefits. The system analyzes formulary tiers, cost-sharing information, prescribing patterns, and medication history to identify opportunities related to high-cost drugs, medication adherence, prior authorization, and step-therapy requirements. It addresses the challenge of balancing medication affordability, appropriate utilization, and patient outcomes.

UC07 – Avoidable Emergency Department Utilization Navigator

This use case focuses on identifying patterns of potentially avoidable emergency-department utilization. The system analyzes historical utilization patterns and identifies cases where a lower-acuity option such as primary care, urgent care, telehealth, or care-management follow-up might have been appropriate. However, the system must be designed carefully so that it never discourages patients from seeking emergency care when it is genuinely necessary.

UC08 – Value-Based Care Contract Performance Command Center

This use case focuses on helping payers evaluate the performance of providers or Accountable Care Organizations under value-based care contracts. The system combines cost, quality, utilization, and provider performance information to identify the factors driving financial gains or losses and recommend actions that could improve contract outcomes.

UC09 – SDOH-Driven Member Risk and Intervention Prioritization

This use case focuses on identifying members who may be at higher risk because of social determinants of health, such as food access, transportation, housing, and income-related factors. The system combines synthetic clinical or utilization information with community-level social-risk indicators to identify vulnerable populations and recommend realistic interventions.

UC10 – Claims and Authorization Data-Quality Anomaly Monitor

This use case focuses on detecting problems in healthcare data pipelines. The system analyzes claims, pharmacy, or authorization data to identify data-quality issues, processing anomalies, and potential SLA risks, while also explaining the likely causes and suggesting corrective actions. The goal is to prevent poor-quality data from affecting downstream payer analytics and healthcare decisions.

UC11 – Multi-Agent Invoice and Expense Exception-Handling Workflow

This use case focuses on automating invoice and expense processing using multiple AI agents. The system extracts information from invoices or receipts, compares it with purchase orders or expense policies, identifies exceptions such as mismatched amounts or missing purchase orders, and routes the transaction either for automatic approval or human review. It is a strong example of agentic AI but is focused on finance operations rather than healthcare.

UC12 – Multi-Agent Research-to-Report Analyst Assistant

This use case focuses on building a multi-agent research workflow. One agent plans the research task, another retrieves evidence from public sources, another synthesizes the information into a structured report, and a reviewer agent checks for unsupported claims or missing evidence before human approval. It is a strong demonstration of agentic AI, but because it is cross-industry, the team would need to define a specific research domain to make the solution sufficiently focused.

Why We Chose UC02 – Prior Authorization

We chose **Prior Authorization Triage and Policy Companion** because it provides the strongest combination of **real-world healthcare impact, clear business requirements, available synthetic patient data, insurance-policy reasoning, machine learning, RAG, and agentic AI**. The problem is also easy to understand: a healthcare provider submits a request, and the insurance organization needs to determine whether that request satisfies the patient's plan coverage and medical-necessity requirements.

Another major reason is that UC02 allows us to use AI technologies for **meaningful purposes rather than adding AI simply for demonstration**. RAG can retrieve the relevant coverage policy, agents can analyze patient and document evidence, and the ML model can perform the final triage recommendation. This creates a complete and explainable workflow where each technology has a specific responsibility.

UC02 also provides a strong **human-in-the-loop** approach. Instead of trying to replace nurses or insurance reviewers, our system can automatically process straightforward cases while routing uncertain or conflicting cases to a nurse for review and identifying requests where additional information is required. This makes the solution more realistic, safer, and easier to justify.

Most importantly, UC02 gives our team an opportunity to demonstrate an end-to-end healthcare AI solution: **patient data, insurance coverage, policy retrieval, clinical evidence, document analysis, agentic reasoning, machine learning, explainability, and human review** all come together around one clearly defined problem. That makes it a stronger choice for demonstrating both technical capability and meaningful real-world impact.

PROJECT DOCUMENTATION

1. Project Overview

- **Project Name:** ProAuth IQ
- **Description:** AI-powered prior authorization triage and policy companion for policy-aware, evidence-based authorization recommendations.
- **Domain:** Healthcare / Insurance / AI
- **Type:** Web + AI/ML
- **Status:** Prototype / Development

2. Team Information

Name	Role	Responsibilities	EMAIL-ID
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3. Problem Statement

Today, when a healthcare provider requests authorization for a medical service, insurance teams often rely on manual review of patient records, clinical information, supporting documents, requested procedures, and complex insurance coverage policies to determine whether the service meets the member's plan requirements.

The challenge becomes more difficult when the required information is spread across different systems and policy documents. Reviewers must identify the correct insurance plan and applicable coverage policy, understand the policy's medical-necessity criteria, verify the patient's clinical evidence, and determine whether the submitted documentation supports the request.

Who experiences this problem?

The problem affects multiple stakeholders:

- Doctors and healthcare providers — spend time preparing authorization requests and responding to additional-information requests.
- Insurance companies and utilization-review teams — handle large volumes of requests that require policy and clinical review.
- Nurses/reviewers — spend time manually reviewing requests that could potentially be triaged automatically.
- Patients/members — may experience delays in receiving treatments or services while authorization is being processed.

Why is the problem important?

Prior authorization requires information from multiple sources: patient clinical records, requested procedures, insurance plans, coverage policies, and supporting documents. Manually connecting these pieces can be time-consuming and inconsistent.

A policy-aware automated system can:

- Reduce repetitive administrative work.
- Quickly identify whether required information is available.
- Retrieve the relevant insurance coverage criteria.
- Match patient evidence against policy requirements.
- Route straightforward requests for faster processing.
- Send uncertain cases to human reviewers.
- Clearly explain the evidence behind each recommendation.

What happens if the problem is not solved?

If prior authorization remains heavily manual and fragmented:

- Providers spend more time on administrative work.
- Insurance reviewers face higher workloads.
- Requests may take longer to process.
- Missing information can lead to repeated submissions.
- Patients may experience delays in receiving appropriate care.
- Lack of transparent reasoning can make the authorization process difficult for providers and reviewers to understand.

ProAuth IQ focuses on making prior authorization **faster, policy-aware, evidence-grounded, transparent, and human-reviewable**, rather than replacing clinical or insurance professionals.

4. Proposed Solution

ProAuth IQ is a policy-aware prior authorization assistant that automatically analyzes an authorization request by combining the patient's clinical information, insurance plan, coverage policy, and submitted documents. It uses RAG to retrieve the relevant policy requirements and AI agents to compare those requirements with the available clinical evidence.

How does the solution solve the identified problem?

Instead of manually checking multiple sources, the system creates a single automated workflow:

Prior Authorization Request → Identify Insurance Policy → Retrieve Relevant Coverage Criteria → Analyze Patient & Documents → Compare Against Policy → ML Triage

The system then recommends one of three outcomes:

- APPROVE — requirements are sufficiently satisfied.
- PEND FOR NURSE REVIEW — the case requires human evaluation due to uncertainty or conflicting evidence.
- REQUEST MORE INFORMATION — required evidence or documentation is missing.

What makes the solution useful or different?

Unlike a simple LLM-based system, ProAuth IQ does not ask an AI model to directly decide whether a request should be approved.

It combines : Policy RAG + Agentic Evidence Analysis + Deterministic Evaluation + ML Triage

This makes the recommendation:

- Policy-aware — based on the applicable insurance coverage criteria.
- Evidence-grounded — decisions are linked to retrieved policy and patient evidence.
- Explainable — reviewers can see why a request was approved, pended, or returned for information.
- Human-in-the-loop — uncertain cases are routed to nurse review rather than forcing an automated decision.
- Configurable — policies and coverage criteria can be updated without rewriting the entire decision system.

5. Objectives

1. Automate Prior Authorization Triage

Develop an intelligent system that analyzes incoming prior authorization requests and recommends **Approve, Pend for Nurse Review, or Request More Information**, reducing repetitive manual work for insurance review teams.

2. Perform Policy-Aware Coverage Evaluation

Identify the patient's **insurance plan and applicable coverage policy**, then evaluate whether the requested medical service and diagnosis satisfy the plan's coverage and medical-necessity requirements.

3. Ground Decisions in Clinical and Policy Evidence

Use patient records, submitted documents, and **RAG-based policy retrieval** to ensure that recommendations are based on relevant evidence rather than unsupported assumptions or generic AI responses.

4. Provide Explainable and Transparent Recommendations

Clearly show which policy criteria were satisfied, what evidence supports them, what information is missing, and why the request was routed to approval, nurse review, or additional-information processing.

5. Enable Intelligent Human-in-the-Loop Decision Support

Automatically handle well-supported requests while identifying uncertain, incomplete, or conflicting cases for **nurse/reviewer intervention**, ensuring that AI assists healthcare professionals rather than replacing human judgment.

6. Build a Policy-Driven ML Triage Model

Develop a machine-learning model that uses structured features from policy evaluation, clinical evidence, documentation completeness, and request information to predict the appropriate triage category.

6. Target Users / Stakeholders

User / Stakeholder	Needs	How They Use the System	Expected Benefit
Doctor / Healthcare Provide	Simple request submission and visibility into missing information	Enters patient, diagnosis, requested service, insurance details, clinical justification, and uploads supporting documents	Faster and easier submission with fewer incomplete requests
Nurse / Manual Reviewer	Clear evidence and reasoning for complex cases	Reviews requests routed for manual review, checks policy criteria, clinical evidence, documents, and system reasoning	Reduces review effort and helps focus on uncertain cases
Patient / Member	Visibility into authorization progress and outcome	Uses a read-only dashboard to view request status and final outcome	Better transparency and reduced uncertainty
Admin	System and policy management	Manages users, roles, insurance plans, policy configurations, and audit records	Better system control, security, and maintainability

7. Scope

7.1 In Scope

- **Prior Authorization Submission** – Doctors can submit a request with patient details, diagnosis, requested medical service, insurance plan, clinical justification, and supporting documents.
- **Patient & Insurance Identification** – The system associates the request with the appropriate patient and insurance plan and identifies the relevant coverage policy.
- **Policy & Coverage Evaluation** – The system checks the requested service and diagnosis against the applicable insurance policy and medical-necessity requirements.
- **Policy RAG** – Relevant sections from insurance/medical-policy documents are retrieved to provide evidence for the coverage evaluation.
- **Agentic Analysis** – Specialized AI agents analyze the policy, patient clinical information, and submitted documents before producing structured evidence.
- **ML-Based Triage** – A machine-learning model evaluates the structured information and predicts **Approve, Pend for Nurse Review, or Request More Information**.
- **Explainable Results** – The system displays the applicable policy, satisfied criteria, missing information, evidence, scores, and reasoning behind the recommendation.
- **Role-Based Dashboards** – Separate interfaces are provided for doctors, nurses, patients, and insurance/admin users based on their responsibilities.
- **Audit Trail** – Authorization requests, policy evidence, evaluation results, ML predictions, and workflow activities are stored for traceability.

7.2 Out of Scope

- **Real Patient Data –** The prototype uses synthetic healthcare data and does not process real patient records.
- **Real Insurance Transactions –** The system does not submit requests to or process transactions through real insurance companies.
- **Autonomous Medical Diagnosis or Treatment –** The system evaluates authorization requirements but does not diagnose patients or recommend treatments.
- **Fraud Detection & Appeal Prediction –** Fraud investigation and appeal-probability prediction are not part of the current version.
- **Real-Time Eligibility Verification –** The prototype does not connect to live insurance systems to verify eligibility.
- **Production EMR/EHR Integration –** Full integration with hospital EMR/EHR systems is outside the current prototype.
- **Protected CMS API Integration –** The prototype will not depend on CMS APIs requiring credentials that are unavailable to the team; publicly available policy documents will be used for the RAG layer.
- **Document or Medical Information Manipulation –** The system will not alter, fabricate, or generate false clinical information or supporting documents.
- **Complete Nationwide Policy Coverage –** The prototype will demonstrate the workflow using a limited, configurable set of policy documents rather than attempting to represent every insurance policy.

8. Functional Requirements

ID	Requirement	User	Priority	Status
FR-01	Create and submit a prior authorization request with patient, diagnosis, service, insurance, clinical details, and documents.	Doctor	High	Planned
FR-02	Retrieve and associate the patient's insurance plan and applicable coverage policy with the request.	System	High	Planned
FR-03	Retrieve relevant coverage and medical-necessity information from policy documents using RAG.	System	High	Planned
FR-04	Analyze patient clinical information and supporting documents to identify relevant evidence and missing information.	System / AI Agents	High	Planned
FR-05	Evaluate the authorization request against applicable policy criteria and generate structured evaluation results.	System / AI Agents	High	Planned

FR-06	Generate policy compliance and evidence-completeness scores based on the evaluated criteria.	System	High	Planned
FR-07	Predict the authorization triage outcome as Approve, Pend for Nurse Review, or Request More Information using the ML model.	System / ML Model	High	Planned
FR-08	Display the recommendation with policy evidence, satisfied criteria, missing information, and reasoning.	Doctor / Nurse / Insurance Agent	High	Planned
FR-09	Allow nurses to view and review requests specifically routed for nurse review.	Nurse	High	Planned
FR-10	Allow patients to view their authorization status and outcome through a read-only dashboard.	Patient	Medium	Planned
FR-11	Allow insurance/admin users to monitor authorization requests and manage relevant policies/configurations.	Insurance Agent / Admin	High	Planned
FR-12	Implement role-based access so users can access only the functions appropriate to their role.	All Users	High	Planned

9. Non-Functional Requirements

Category	Requirement	Target / Standard	Priority
Performance	Fast request processing	Triage response within 5 seconds	High
Security	Secure and role-based access	Authentication, role-based permissions, and encrypted communication	High
Scalability	Extensible architecture	Support additional users, policies, agents, and ML models without major redesign	High
Usability	Simple and clear interface	Easy request submission with role-specific dashboards and understandable results	High
Availability	Reliable system operation	Prevent data loss and handle component/service failures gracefully	High
Accessibility	Accessible interface	Clear labels, readable content, sufficient contrast, and keyboard-friendly navigation	Medium

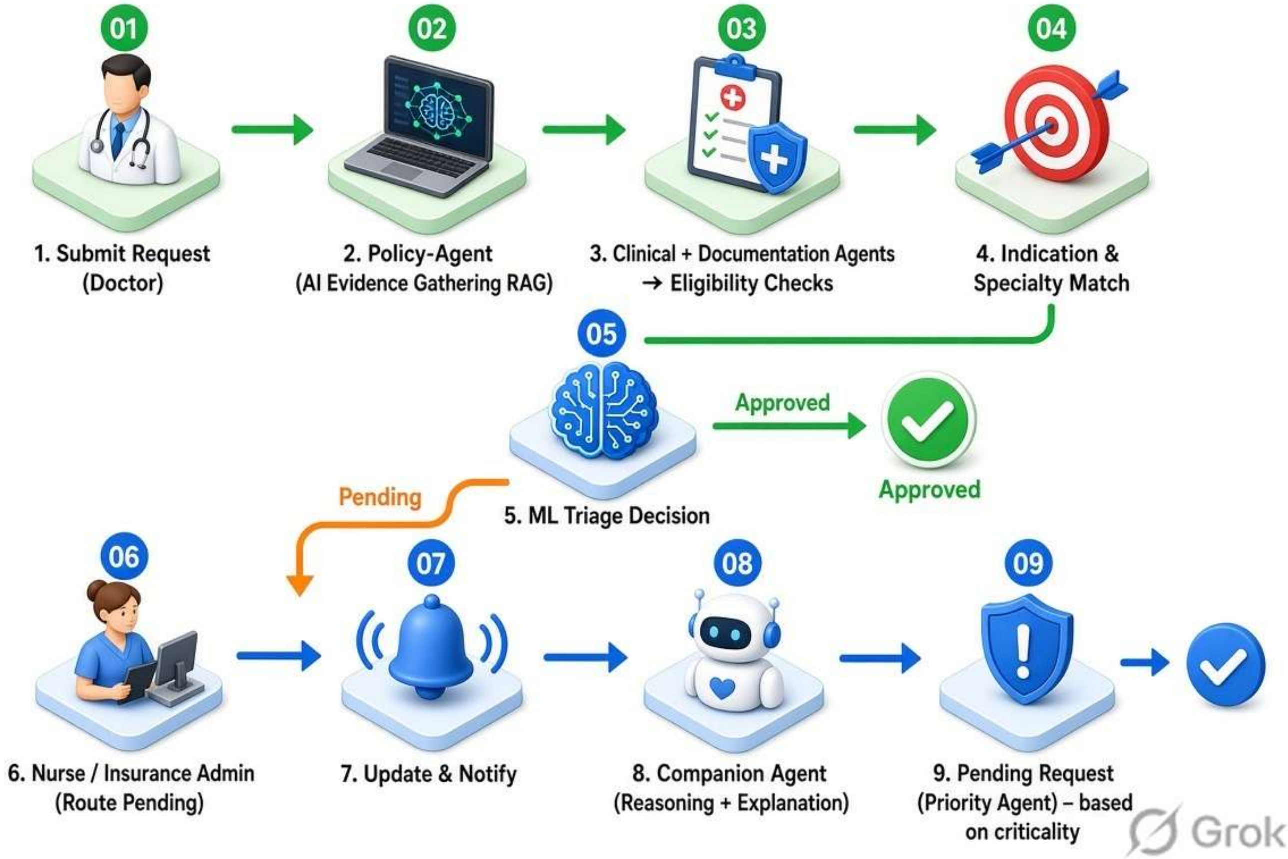
10. Features

Feature	Description	Input	Output	Status
Authorization Request	Doctor creates and submits a prior authorization request	Patient, diagnosis, service, insurance, clinical details	Submitted request	Planned
Document Upload & Analysis	Upload and analyze supporting clinical documents	Medical documents + request context	Relevant evidence / missing information	Planned
Policy Identification	Identify the applicable Medicare coverage policy	Insurance plan, diagnosis, service	Applicable policy	Planned
Policy RAG	Retrieve relevant coverage and medical-necessity criteria from policy documents	Policy documents + request context	Relevant policy evidence	Planned
Clinical Evidence Analysis	Analyze patient information relevant to the requested service	Patient clinical data	Clinical evidence	Planned
Policy & Coverage Evaluation	Compare the request and evidence against the retrieved policy criteria	Policy + clinical/document evidence	Criteria results & compliance score	Planned
ML-Based Triage	Predict the appropriate authorization workflow outcome	Structured evaluation features	Approve / Nurse Review / More Information	Planned
Explainable Recommendation	Provide evidence and reasoning behind the predicted outcome	Policy, agent and ML results	Recommendation, evidence, missing criteria	Planned
Role-Based Dashboards	Provide relevant views for doctors, nurses, patients and insurance/admin users	User role + authorization data	Role-specific information and actions	Planned
Audit Trail	Track authorization processing and system decisions	Requests, policies, evaluations, predictions, actions	Traceable authorization history	Planned

11. User Roles & Permissions

Role	Access	Can Create	Can Edit	Can Delete
Doctor	Create and view own authorization requests	Authorization Request, Documents	Own draft requests before submission	Own drafts
Nurse	View and review requests routed for nurse review	Review notes/status where applicable	Review information	No
Patient	Read-only access to own requests	No	No	No
Admin	Full system administration	Users, policies, configurations	Users, policies, configurations	Users/configuration records

12. User Flow / System Workflow



13. UI/UX Documentation

13.1 Design Goals

- Provide a **simple and intuitive user experience**.
- Use a professional **healthcare and insurance-oriented visual style**.
- Provide **role-based dashboards** for doctors, nurses, patients, and insurance/admin users.
- Clearly display authorization **status, policy criteria, evidence, missing information, and ML confidence**.
- Reduce unnecessary navigation and keep important actions easily accessible.
- Present AI-generated results in a **clear and explainable format**.
- Maintain consistent layouts, typography, colors, and components across the application.
- Ensure the interface is responsive and accessible across common screen sizes.

13.2 Screens / Pages

Screen	Purpose	Main Components	User Actions	Status
Login	Authenticate users	Login form, role-based authentication	Login	Completed
Doctor Dashboard	Manage authorization requests	Request summary, status cards, recent requests	View, create, track requests	Completed
Create Authorization	Submit a new prior authorization request	Patient details, diagnosis, service, insurance, clinical information	Enter details and submit	Completed
Document Upload	Add supporting evidence	File uploader, document list, upload status	Upload/remove documents	Completed
Authorization Details	View complete request information	Patient details, service, documents, status	View request	Completed
Policy & Evidence	Display applicable policy and evidence	Policy criteria, retrieved sections, evidence	Review policy and evidence	In Development
AI Evaluation	Display agent-based evaluation	Clinical analysis, document analysis, policy evaluation	View evaluation	In Development
Triage Result	Display ML recommendation	Prediction, confidence, criteria, missing information	View recommendation	In Development
Nurse Review Dashboard	View requests requiring review	Pending requests, priority, evaluation summary	Review case	Completed
Nurse Review Details	Provide detailed review context	Patient information, policy evidence, AI results	Review request	Completed
Patient Dashboard	Track authorization requests	Request list, status, notifications	View status	Completed
Patient Authorization Details	Show request progress	Status, request summary, updates	View details	Completed
Admin Dashboard	Monitor overall authorization workflow	Statistics, request status, analytics	Monitor requests	Completed
Policy Management	Manage available policy information	Policy list, search, policy details	View/manage policies	In Development

Audit Trail	Track authorization activities	Timeline, events, timestamps	View activity history	In Development

14. Technology Stack

Layer	Technology	Version	Purpose
Frontend	React+vite	React: 19.1.0 Vite: 7.0.0	Provides the interactive user interface and role-based dashboards for doctors, nurses, patients, and insurance/admin users.
Backend	Node.js+Express	Node.js: 24.19.0 FASTAPI: 0.115.6	Handles REST APIs, authentication, authorization requests, business logic, and communication with the database and AI services.
Database	PostgreSQL	PostgreSQL 18	Stores users, patients, authorization requests, insurance policies, evaluations, predictions, documents, and audit records.
AI / ML	Python and FAST-API	Python: 3.12.3 FASTAPI: 0.115.6	Hosts the machine-learning model and exposes prediction APIs for prior authorization triage.
RAG and Agents	Python and Lang-Graph	Python: 3.12.3 Lang-Graph: 1.2.11 python-dotenv: 1.0.1 sentence-transformers: 3.3.1 pytesseract: 0.3.13	Orchestrates AI agents for policy retrieval, clinical/document analysis, evidence evaluation, and workflow reasoning.
API	REST-API and FAST-API	FASTAPI: 0.115.6 RESTAPI: /api/v1	Enables structured communication between the frontend, backend, ML model, and agent services.
Hosting	AWS-Cloud	AWS CLI v2	Provides the cloud environment for deploying and running the application components.
LLM	Groq-LLM	Groq: 0.13.1	Provides LLM-based reasoning for policy, clinical, and document analysis within the agent workflow.

15. System Architecture

ProAuth IQ follows a **layered agentic architecture** where the authorization request moves from the user portals through request management, AI agents, policy/data sources, and ML-based triage.

1. **Application Portals –** Doctors submit prior authorization requests, while nurses, admins, and patients access information relevant to their roles.

2. **Request & Data Layer –** The system manages the submitted authorization request, uploaded documents, and structured patient/request data before sending them for analysis.

3. **Agentic Layer** – A **LangGraph-based orchestrator** coordinates specialized agents:

    - **Policy Agent –** identifies and retrieves relevant coverage-policy information.

    - **Clinical Agent –** analyzes the clinical context of the request.

    - **Document Analysis Agent –** examines submitted documents and identifies relevant or missing evidence.

    - **Coverage Reasoning Agent –** combines the available evidence and evaluates the request against the applicable coverage requirements.

4. **Data Storage Layer – PostgreSQL** stores structured application data, document storage maintains submitted files, and a **vector database** supports policy/document retrieval for the RAG pipeline.

5. **ML Triage Layer –** The evidence and structured features generated by the agentic layer are passed to the ML model, which classifies the request into the appropriate triage outcome:

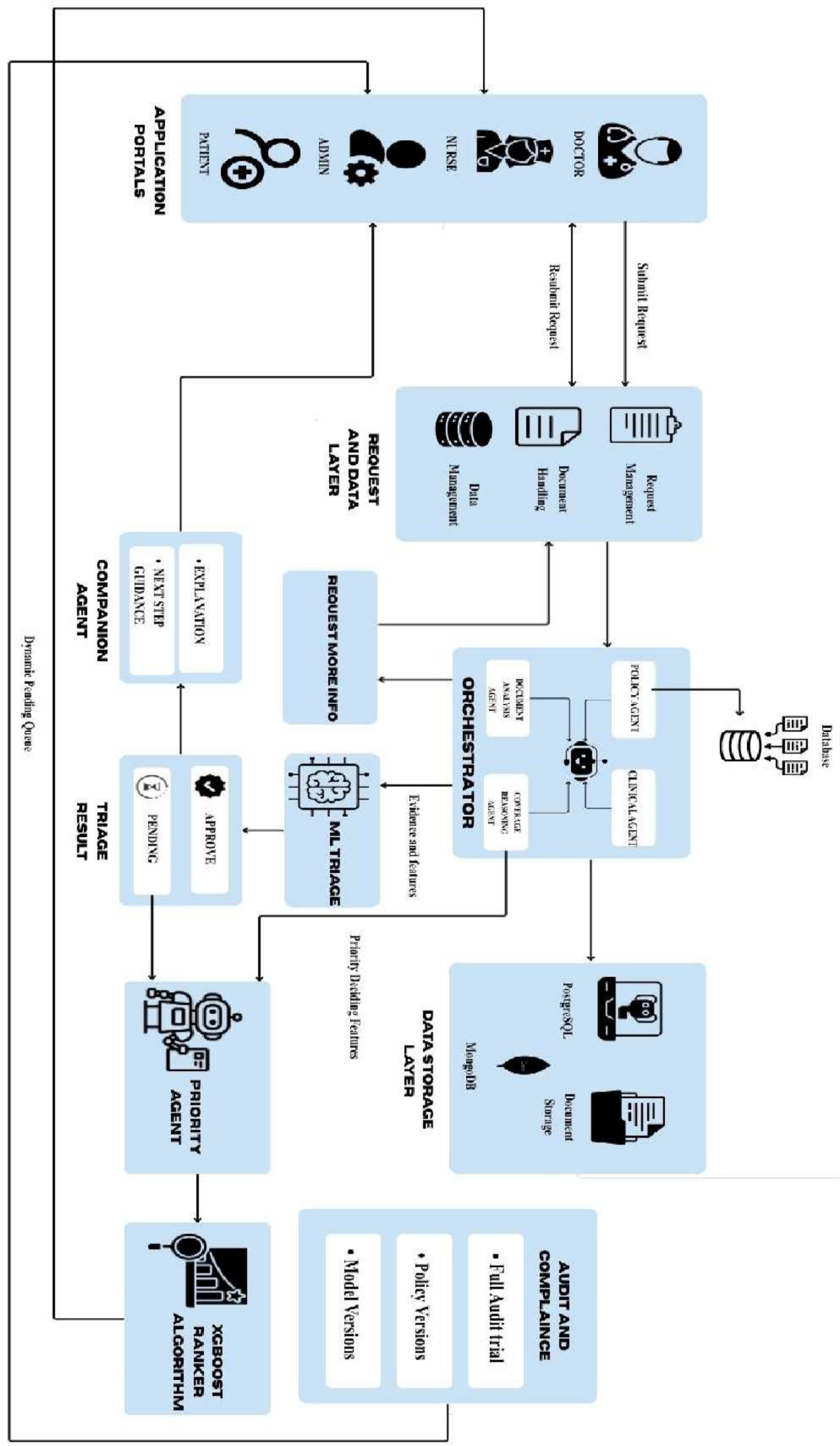
    - **Approve**

    - **Pending / Nurse Review**

    - **Request More Information**

6. **Companion Agent –** Provides an understandable explanation of the result and guides the user on the appropriate next step, particularly when additional information is required.

7. **Audit & Compliance Layer –** Records the authorization workflow, policy versions, model versions, and other important activities to provide a complete audit trail.



System Architecture

16. Database Design

Database Type: PostgreSQL relational database was selected because the system contains structured and related data such as patients, medical records, payers, policies, authorization requests, and decisions. A relational database allows these entities to be connected through primary and foreign keys and supports consistent querying and data management.

Table / Collection	**Purpose**	**Important Fields**	**Relationships**
Patients	Stores patient information	patient_id, name, date_of_birth, gender, contact_details	Connected to Medical Records, Payers and Authorization Requests
Medical_Records	Stores patient's medical information	record_id, patient_id, diagnosis, symptoms, medical_history	patient_id → Patients
Diagnoses	Stores diagnosed conditions	diagnosis_id, patient_id, diagnosis_code, diagnosis_name, date	patient_id → Patients
Procedures	Stores requested/performed medical procedures	procedure_id, patient_id, procedure_code, procedure_name, date	patient_id → Patients
Medications	Stores prescribed medications	medication_id, patient_id, medicine_name, dosage, frequency	patient_id → Patients
Payers	Stores insurance companies/programs	payer_id, payer_name, payer_type	Connected to Patients, Plans and Claims
Insurance_Plans	Stores specific insurance plans	plan_id, payer_id, plan_name, plan_type, state	payer_id → Payers
Policies	Stores healthcare policy information	policy_id, plan_id, policy_name, policy_type, effective_date, source_url	plan_id → Insurance_Plans
Coverage_Rules	Stores policy coverage criteria	rule_id, policy_id, procedure_code, diagnosis_code, coverage_status	policy_id → Policies
Medical_Necessity	Stores medical-necessity requirements	necessity_id, rule_id, criteria, conditions, documentation_required	rule_id → Coverage_Rules
Prior_Authorizations	Stores authorization requests	authorization_id, patient_id, procedure_id, payer_id, policy_id, status	Connected to Patients, Procedures, Payers and Policies
Authorization_Documents	Stores documents submitted for authorization	document_id, authorization_id, document_type, document_path, uploaded_date	authorization_id → Prior_Authorizations
Claims	Stores insurance claim information	claim_id, patient_id, payer_id, procedure_id, claim_amount,	Connected to Patients, Payers and Procedures

		claim_status	
AI_Decisions	Stores AI-generated authorization decisions	decision_id, authorization_id, decision, confidence_score, reason	authorization_id → Prior_Authorizations
Policy_Sources	Stores the source of policy information	source_id, policy_id, source_name, source_url, retrieved_date	policy_id → Policies

17. API Documentation

Method	Endpoint	Purpose	Request	Response	Auth	Status
POST	`/api/v1/auth/login`	Authenticate users	Email, password	JWT token, role	No	Planned
GET	`/api/v1/patients/:id`	Retrieve patient information	Patient ID	Patient details	JWT	Planned
POST	`/api/v1/authorizations`	Create prior authorization request	Patient, diagnosis, service, insurance, clinical details	Authorization ID	Doctor	Planned
GET	`/api/v1/authorizations/:id`	View authorization details	Authorization ID	Request details	JWT	Planned
PUT	`/api/v1/authorizations/:id`	Update authorization request	Updated request data	Updated request	Doctor	Planned
POST	`/api/v1/authorizations/:id/documents`	Upload supporting documents	Medical document	Document status	Doctor	Planned
POST	`/api/v1/authorizations/:id/submit`	Submit request for evaluation	Authorization ID	Submission status	Doctor	Planned
POST	`/api/v1/authorizations/:id/evaluate`	Evaluate request against policy and evidence	Authorization ID	Evaluation results	JWT	Planned
GET	`/api/v1/authorizations/:id/evaluation`	Retrieve evaluation details	Authorization ID	Criteria, evidence, missing information	JWT	Planned
POST	`/api/v1/authorizations/:id/triage`	Generate ML triage prediction	Evaluation features	Prediction, confidence	JWT	Planned
GET	`/api/v1/authorizations/:id/status`	Retrieve authorization status	Authorization ID	Current status	JWT	Planned
GET	`/api/v1/reviews/pending`	Retrieve requests	Filters	Pending requests	Nurse	Planned

		requiring nurse review				
GET	`/api/v1/patient/authorizations`	View patient's requests	—	Authorization history	Patient	Planned
GET	`/api/v1/policies`	Retrieve available Medicare policies	Search/filter	Policy list	Admin	Planned
POST	`/api/v1/policies/retrieve`	Retrieve relevant policy evidence using RAG	Diagnosis, service, plan	Relevant policy sections	System	Planned
GET	`/api/v1/authorizations/:id/audit`	Retrieve authorization audit trail	Authorization ID	Audit history	JWT	Planned

18. AI / ML Documentation (if applicable)

Model 1 — Approval Decision Model

Problem it solves: For a prior authorization request that has enough evidence to be reviewed, decide whether it should be Approved or Pended.

Dataset: 14,000 synthetic prior authorization requests, covering 9 Medicare service categories (MRI scans, CPAP therapy, wheelchairs, bariatric surgery, etc.), built using real Medicare coverage rules for what each service requires. The data was split by patient into training, validation, and test sets, and one entire service category was kept completely separate to test the model on a type of request it had never seen.

Preprocessing: The pipeline originally produced 10 pieces of information about each request. Testing showed 6 of these were duplicates of each other in disguise — mathematically the same information written differently — so they were removed. The remaining 4 were checked and confirmed as the right set to use: policy fit score, documentation completeness score, clinical evidence score, and number of missing/unclear items. The values were scaled to a common range, and each data split was balanced so the model saw an equal number of Approve and Pend examples.

Model used: Logistic Regression. This model calculates a score for each request and converts it into a probability of approval. It was compared against a Random Forest model and performed equally well, so the simpler, fully explainable model was kept.

Training process: The model was trained on the training data using cross-validation to pick the best settings. The cutoff point for deciding Approve vs. Pend was then chosen using the validation data only, set to specifically reduce wrongful approvals — a wrongful approval is treated as three times more costly than a wrongful pend. The model was then retrained on the training and validation data combined, and tested once on the test data and once on the unseen service category.

Evaluation results:

Test data accuracy: 83.7%

Unseen service category accuracy: 81.5%

Both results were also measured with ROC-AUC, a score that checks how well the model separates approvable from non-approvable cases: 0.93 on test, 0.91 on the unseen category.

Output: A probability between 0 and 1 that the request should be approved, and a final Approve/Pend decision based on the cutoff point.

Explainability: The decision is a simple formula — each of the 4 pieces of information is multiplied by a fixed weight, added together, and converted to a probability. Every weight can be read directly: all three score-based features push toward approval, and the missing-items count pushes toward pending.

Limitations: The model was trained on synthetic data, not real historical approval records, so its real-world accuracy is not yet proven. It assumes each of the 4 factors adds up in a straight line rather than interacting in more complex ways. The cutoff point was deliberately set to catch more wrongful approvals, which means some approvable requests get pended unnecessarily. This model does not make any decision about requests missing too much information — those are filtered out before this model ever sees them.

Model 2 — Pending Case Priority Model

Problem it solves: Once a request is marked Pending, someone still has to review it. This model decides the order in which pending cases should be reviewed — it does not approve, deny, or pend anything itself.

Dataset: 3,185 requests that Model 1 marked as Pending, covering about 2,000 patients. There is no historical record of the order nurses actually reviewed cases in, so the training labels were built from a written prioritization rule instead: safety concerns rank highest, followed by long-waiting cases and high-urgency cases, followed by cases with incomplete evidence, with routine cases ranked lowest. Some random variation was added so the model learns general patterns rather than the rule word-for-word.

Preprocessing: Each pending case was described using the same 3 score features from Model 1, plus new information specific to timing and urgency: how long the case has been waiting, how close it is to its review deadline, its stated urgency level (routine, urgent, emergency), and whether any safety concern was flagged. Cases were grouped into batches so the model could learn which cases in the same batch should come before others, rather than scoring each case in isolation.

Model used: XGBoost Ranker, a model built specifically to learn ordering rather than yes/no answers. This is a different type of model from Model 1 because the question being answered is different — Model 1 answers "what is the decision," Model 2 answers "what order should these go in."

Training process: The model was trained on the pending-case data, split into training, validation, and test groups, with each batch of cases kept whole within one split.

Evaluation results: Since this model produces an order rather than a decision, it is not measured with accuracy. It is measured with a ranking score called NDCG, which checks whether the cases the model puts near the top of the queue are actually the ones that should be near the top, based on the priority rule it was trained on:

Test data score: 98.3%

Output: A priority score for each pending case, combined with the model's ranking to produce a final position in the queue. If a case has an emergency flag, has already missed its review deadline, or is marked as a critical safety concern, it is automatically placed at the top of the queue regardless of what the model's own score says. Every case also gets a short written reason explaining why it was ranked where it is. The queue is recalculated regularly, so a case that has been waiting longer moves up over time.

Explainability: Every position in the queue has a plain-language reason attached to it, so a reviewer can see exactly why one case outranks another. Safety and deadline rules always take priority over the model's own score — the model can only reorder cases within what those rules allow.

Limitations: The 98.3% score means the model closely matches the written rule it was trained on — it does not mean the model matches how real nurses would actually prioritize cases, because that real-world data does not exist yet. This model never makes a medical or coverage decision; its only role is ordering an already-existing list of pending cases.

19. Development Setup

1. **Prerequisites**

- Node.js 24.19.0
- Npm
- Python 3.12.3
- PostgreSQL 18
- Git
- Docker and Docker Compose
- AWS CLI v2
- Groq API Key

2. **Installation Commands**

- Frontend and Backend:
- npm install
- AI/ML Services:
- pip install -r requirements.txt
- Database:
- docker compose up -d

3. **Environment Variables**

- Create a .env file and configure:
- PORT=5000
- NODE_ENV=development
- DATABASE_URL=postgresql://<user>:<password>@localhost:<port>/<database>
- JWT_SECRET=<your-secret>
- JWT_EXPIRES_IN=1d
- GROQ_API_KEY=<your-groq-api-key>
- ML_SERVICE_URL=http://localhost:8001

- STORAGE_PROVIDER=local
- API keys, passwords, and other sensitive credentials should not be committed to the source repository.

4. **How to Run Frontend**

- npm run dev
- This starts the React application using Vite.

5. **How to Run Backend**

- npm run dev
- This starts the Node.js and Express backend in development mode.
- For production: npm start

6. **How to Run Database**

- Start PostgreSQL using Docker Compose: docker compose up -d
- Verify the running containers: docker ps

7. **How to Run AI/ML Service**

- uvicorn main:app --reload --port 8001
- This starts the Python FastAPI service used by the ML and AI components.

8. **How to Build for Production**

- Frontend:
  npm run build
  The production frontend is generated in the dist directory.
- Backend:
  npm install --omit=dev
  npm start
  The AI/ML service is deployed separately as a FastAPI service.

20. Project Folder Structure

```
├── ProAuth_AI_FrontEnd/
│   └── React / Vite Application
│       ├── Doctor
│       ├── Nurse
│       ├── Insurance Admin
│       └── Patient
├── ProAuth_AI_BackEnd/
│   └── Node.js / Express API
│       ├── Authentication
│       ├── Authorizations
│       ├── Documents
│       ├── Reviews
│       ├── Queue
│       ├── Analytics
│       └── Audit
├── ProAuth_AI_ML/
│   ├── policy-rag/
│   │   ├── agents/
│   │   │   ├── Policy Agent
│   │   │   ├── Clinical Agent
│   │   │   ├── Document Agent
│   │   │   ├── Coverage Reasoning
│   │   │   └── Companion Agent
│   │   ├── rag/
│   │   │   └── pgvector retrieval
│   │   ├── ml/
│   │   │   └── Logistic Regression model
│   │   └── triage_graph.py
│   ├── priority_intelligence/
│   │   └── agent/
│   │       └── Priority LangGraph
│   │   ranker.py
│   │   safety.py
│   └── tiers.py
├── copilot/
│   └── AWS service manifests
├── iam/
│   └── IAM policies
└── DEPLOYMENT.md
```

21. Testing

Test ID	Feature	Test Case	Expected Result	Actual Result	Status
TC-001	Login	Enter valid user credentials	User successfully logs in	Successful login	Pass
TC-002	Authentication	Enter invalid credentials	Login is rejected with error message	Error displayed	Pass
TC-003	Authorization Creation	Submit valid patient and service details	Authorization request is created	Request created	Pass
TC-004	Required Fields	Submit request with missing mandatory fields	Validation message is displayed	Validation displayed	Pass
TC-005	Document Upload	Upload valid supporting document	Document is uploaded successfully	Upload successful	Pass
TC-006	Document Validation	Upload unsupported file type	File is rejected	File rejected	Pass
TC-007	Authorization Retrieval	Open existing authorization	Correct request details are displayed	Details displayed	Pass
TC-008	Database Integration	Create and retrieve authorization	Data is correctly stored and retrieved	Data retrieved correctly	Pass
TC-009	Policy Retrieval	Submit request with diagnosis and service	Relevant policy information is retrieved	Policy retrieved	Pass
TC-010	RAG Evaluation	Search for relevant policy criteria	Relevant policy sections are returned	Results generated	In Testing
TC-011	Clinical Analysis	Provide diagnosis and clinical context	Relevant clinical evidence is identified	Analysis generated	In Testing
TC-012	Document Analysis	Submit clinical document	Relevant and missing evidence is identified	Analysis generated	In Testing
TC-013	Agent Integration	Run complete LangGraph workflow	Agents execute in the expected sequence	Workflow executed	In Testing
TC-014	Policy Grounding	Evaluate request against retrieved policy	Evaluation is based on retrieved	Evaluation generated	In Testing

			evidence		
TC-015	ML Integration	Send structured features to ML API	ML model returns a triage prediction	Prediction returned	In Testing
TC-016	Triage – Approve	Submit request satisfying required criteria	Request is classified as Approve	Prediction generated	In Testing
TC-017	Triage – Nurse Review	Submit uncertain/complex request	Request is classified as Pend for Nurse Review	Prediction generated	In Testing
TC-018	Triage – More Information	Submit request with missing evidence	Request is classified as Request More Information	Prediction generated	In Testing
TC-019	Confidence Score	Generate ML prediction	Prediction includes confidence score	Score generated	In Testing
TC-020	Explainability	View evaluation result	Policy evidence and missing criteria are displayed	Results displayed	In Testing
TC-021	Nurse Dashboard	Login as nurse	Pending cases are displayed	Cases displayed	Pass
TC-022	Patient Dashboard	Login as patient	Patient's authorization status is displayed	Status displayed	Pass
TC-023	Admin Dashboard	Login as admin	Authorization statistics are displayed	Dashboard displayed	Pass
TC-024	Role Authorization	Access restricted feature with unauthorized role	Access is denied	Access denied	Pass
TC-025	Audit Trail	Complete authorization workflow	Important activities are recorded	Audit events recorded	In Testing

22. Challenges & Solutions

Challenge	Cause	Solution	Result
Complex coverage policies	Policies contain lengthy and detailed coverage criteria	Implemented **Policy RAG** to retrieve relevant policy sections	More relevant policy information for evaluation
Hardcoded policy rules	Static rules cannot adapt to changing policy requirements	Move toward **policy-grounded and configurable evaluation**	More flexible policy checking

Unstructured clinical information	Clinical evidence may come from different fields and documents	Use the **Clinical Agent** to structure relevant clinical information	Better evidence extraction
Medical document analysis	Important evidence may be present only in uploaded documents	Use **Document Analysis Agent + OCR/text extraction**	Supporting and missing evidence can be identified
Reliable AI reasoning	LLMs may generate unsupported information	Ground agent reasoning using retrieved policy and submitted evidence	More reliable and traceable reasoning
Agent and ML integration	ML models require structured inputs	Convert agent outputs into structured ML features	Consistent inputs for ML triage

23. Performance / Evaluation

Metric	Target	Actual	Comments
ML Accuracy	≥ 80%	TBD	To be evaluated using test data
Precision	≥ 75%	TBD	Evaluate prediction reliability
Recall	≥ 75%	TBD	Important for identifying requests requiring further action
F1-Score	≥ 75%	TBD	Balance between precision and recall
Triage Confidence	≥ 80% for high-confidence cases	TBD	Low-confidence cases can be routed for Nurse Review
Policy Retrieval Relevance	≥ 85%	TBD	Evaluate whether retrieved policy sections are relevant
Document Analysis Accuracy	≥ 80%	TBD	Evaluate extracted supporting/missing evidence
API Response Time	< 3 seconds for standard API operations	TBD	Excludes longer AI/LLM processing where applicable
System Availability	≥ 99%	TBD	Target for deployed environment
Audit Completeness	100% of critical workflow events	TBD	Requests, evaluations, predictions, and status changes should be traceable

Metric	Target	Actual	Comments
ML Accuracy	≥ 80%	TBD	To be evaluated using test data

24. Deployment

Hosting / Cloud Platform

- **AWS Cloud** is used as the deployment platform.
- Frontend, backend, and AI/ML services can be deployed as separate application services.

Deployment Architecture

- **React + Vite** → Frontend application
- **Node.js + Express** → Backend and REST API
- **PostgreSQL** → Application database
- **Python + FastAPI** → ML/AI service
- **LangGraph + RAG** → Agentic and policy retrieval workflow
- **Groq LLM** → LLM reasoning service

Domain / Application URL

- **Application URL:** To be configured after deployment.
- **API Base URL:** /api/v1

Production Configuration

- Environment variables are used for database credentials, API keys, JWT secrets, LLM configuration, and service URLs.
- CORS, authentication, HTTPS, and production database configuration are enabled for secure communication.

Monitoring / Logging

- Backend API requests and errors are logged.
- AI/ML service execution and prediction errors are monitored.
- Authorization workflow and important system activities are recorded through the audit trail.

25. Future Enhancements

1. **Expand Policy RAG** – Support a larger collection of Medicare policies and eventually additional payer policies.
2. **Improve ML Model** – Train the model using larger and more representative authorization datasets to improve triage performance.
3. **Advanced Document Understanding** – Improve extraction from complex medical documents, scanned PDFs, and different document formats.
4. **FHIR Integration** – Integrate with standardized healthcare data sources and Electronic Health Record systems using FHIR.
5. **Continuous Policy Updates** – Automatically detect and update the RAG knowledge base when new or revised coverage policies become available.

26. Limitations

- The system primarily focuses on **Medicare coverage policies** and does not currently represent all private insurance or payer-specific policies.
- **Synthea provides synthetic patient data**, which may not fully represent the complexity and variability of real-world authorization requests.
- The ML model's performance depends on the **quality, size, and representativeness of the available training data**.
- LLM and agent outputs may still contain errors; therefore, the system uses retrieved evidence and structured outputs to reduce unsupported reasoning.
- The system provides **triage recommendations rather than replacing qualified clinical or insurance professionals**.
- Real-time access to all CMS policy information may be limited by **API availability, licensing requirements, or data update schedules**.
- The current prototype may not include full-scale integration with production EHR, payer, or hospital systems.

27. Daily Development / Work Log

Day	Date	Work Done	Decisions / Changes	Challenges	Result / Output
Day 1	18.08.2026	• Conducted initial research on the healthcare and Prior Authorization domain. • Studied existing healthcare policies, insurance programs, and the Prior Authorization process. • Researched and collected relevant healthcare datasets from available APIs and public sources. • Analyzed the collected datasets to understand the available patient, medical, payer, and healthcare-related information. • Studied RAG	• Decided to use healthcare datasets as the initial source of clinical and payer information. • Decided to use **RAG** so that the AI model can retrieve relevant healthcare policy information before generating a response. • Identified the need to keep clinical data and policy data as separate sources.	• Understanding the healthcare domain and identifying suitable datasets and policy sources. • Understanding how RAG works and how retrieved information can be connected to the AI model.	• Identified suitable datasets and data sources for the project. • Gained an understanding of the healthcare policy data required for the system. • Established the initial approach of using RAG to retrieve relevant policy information before generating the AI response.

		(Retrieval-Augmented Generation) and understood how it can be used to provide relevant policy information to the AI model. • Learned the basic workflow of integrating retrieved healthcare policy documents/data with an AI model.			
Day 2	19.08.2026	• Designed the initial system architecture for the Prior Authorization AI project. • Planned the overall flow between the frontend, backend, database, AI model, and policy data. • Created the initial frontend floor plan, including the main screens, navigation, and placement of major UI components. • Discussed how clinical data, payer information, and healthcare policy data would flow through the system.	• Finalized the initial system architecture and component flow. • Decided on the basic structure and navigation of the frontend. • Decided to keep the clinical data, payer information, and policy data organized as separate components within the system.	• Designing an architecture that properly connects the clinical data, payer information, policy data, RAG system, and AI model. • Deciding how the frontend should present complex healthcare and Prior Authorization information clearly.	• Established the initial system architecture. • Created the frontend floor plan and identified the major UI components /screens. • Established the initial flow between the frontend, backend, database, policy data, and AI components.
Day 3	20.08.2026	• Completed the initial RAG (Retrieval-Augmented Generation) implementation. • Completed ML Model 1 development and integration. • Completed the initial frontend development based on the planned UI floor plan. • Integrated the major	• Finalized the initial RAG approach for retrieving relevant information. Finalized the first ML model and its role in the system. Implemented the planned frontend structure and user interface. • Decided to proceed with	• Ensuring the RAG system retrieves relevant information correctly. • Integrating the ML model with the overall system workflow. • Connecting the completed frontend with the backend/AI components.	• RAG implementation completed. • ML Model 1 completed. • Initial frontend completed. • Major initial components of the system were ready for integration and further testing.

		components developed so far and verified their basic workflow.	integration and testing of the completed components.		
Day 4	21.08.2026	• Completed the AI agent workflow for the Prior Authorization system. • Designed the flow for how the agent processes user requests and interacts with the required system components. • Completed the database implementation. • Structured the database to store and manage the required project information. • Connected the agent workflow and database with the overall system architecture.	• Finalized the agent workflow for handling the Prior Authorization process. • Finalized the database structure and data organization. • Decided how the agent would interact with the database and other system components.	• Designing the agent workflow to handle the required steps in the correct sequence. • Establishing proper communication between the agent and database. • Ensuring the database structure supports the requirements of the system.	• Agent workflow completed. • Database implementation completed. • Agent and database were added to the overall system architecture. • Core backend components were ready for further integration and testing.
Day 5	22.08.2026	• Completed the AI agent development for the Prior Authorization system. • Completed the frontend and backend integration. • Connected the frontend interface with the backend services and AI components. • Tested the communication between the major system components.	• Finalized the AI agent workflow and its integration with the backend. • Connected the planned frontend components with the backend APIs. • Finalized the overall communication flow between the frontend, backend, database, RAG, and AI agent.	• Ensuring smooth communication between the frontend and backend. • Integrating the completed AI agent with the existing system components. • Resolving integration issues between different components.	• AI agent development completed. • Frontend and backend integration completed. • Major system components were connected and ready for end-to-end testing.
Day 6	23.08.2026	• Started the cloud deployment of the project on AWS. • Configured the required AWS environment for deploying the application. • Started deploying the project components to the cloud.	• Decided to deploy the application on AWS for cloud-based accessibility and testing. • Finalized the initial deployment approach and cloud	• Configuring the AWS environment and deploying the application components correctly. • Ensuring the deployed components communicate properly in the cloud. • Organizing the technical information into a clear and	• AWS cloud deployment started. • Initial cloud environment and deployment setup completed. • Project presentation development started.

		• Started preparing the project presentation. • Began organizing the project details, architecture, workflow, technologies, and implementation for the presentation.	environment. • Decided on the structure and major topics to be included in the project presentation.	concise presentation.	• Initial presentation structure and content were prepared.
Day 7	24.08.2026	• Completed the cloud deployment of the application on AWS. • Performed deployment verification and basic maintenance activities. • Completed the project presentation. Completed the project documentation covering the architecture, technologies, workflow, implementation, and project details. • Reviewed the overall project and finalized the required deliverables.	• Finalized the AWS deployment and maintenance approach. • Finalized the presentation structure and content. Finalized the project • documentation and supporting technical details. • Decided to perform a final review of the complete system and project deliverables.	• Ensuring the deployed application and its components were working correctly in the cloud environment. • Organizing the technical implementation and results clearly in the presentation and documentation. • Reviewing the completed project for consistency and completeness.	• AWS deployment completed. • Deployment maintenance and verification completed. • Project presentation completed. • Project documentation completed. • Final project deliverables were prepared and ready for submission/presentation.

28. Project Timeline

Phase	Start Date	End Date	Status
Planning & Research	18.08.2026	19.08.2026	Completed
Frontend	19.08.2026	22.08.2026	Completed
Backend	20.08.2026	22.08.2026	Completed
Database	20.08.2026	21.08.2026	Completed
AI/ML	19.08.2026	22.08.2026	Completed
Testing	22.08.2026	23.08.2026	Completed
Deployment	23.08.2026	24.08.2026	Completed

29. References

1. **CMS Medicare Coverage Database (MCD) –** Official source for Medicare coverage documents, including NCDs, LCDs, Articles, and related coverage information.
2. **CMS Medicare Coverage Database Downloads –** Used as a source for downloadable Medicare coverage datasets and data dictionaries for LCDs, Articles, and NCDs.
3. **Synthea – Synthetic Patient Data –** Used for synthetic patient and healthcare-record data without relying on real patient information. Synthea provides patient records in formats including CSV and FHIR.

30. Final Project Summary

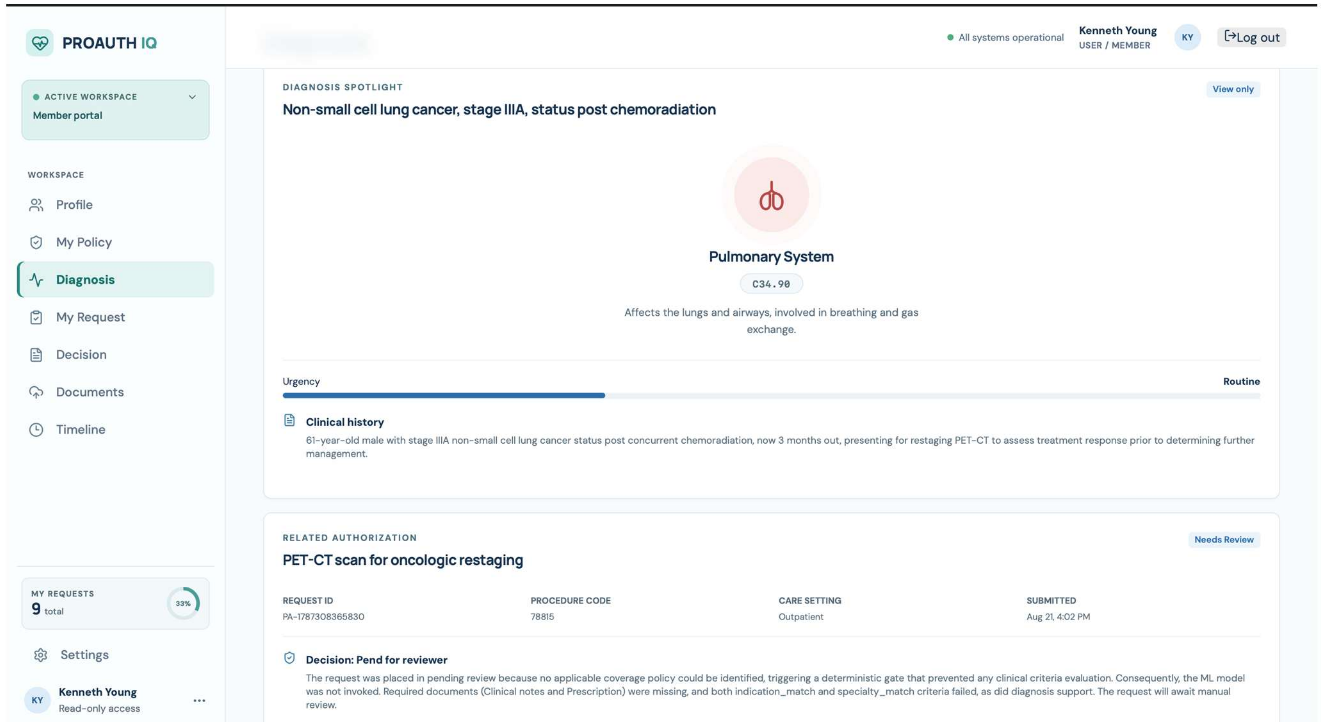
ProAuth IQ is an AI-powered **Prior Authorization Triage and Policy Companion** designed to improve the evaluation of healthcare authorization requests. Prior authorization requires reviewing patient information, clinical evidence, supporting documents, and complex insurance coverage policies, which can make the process time-consuming and difficult to scale.

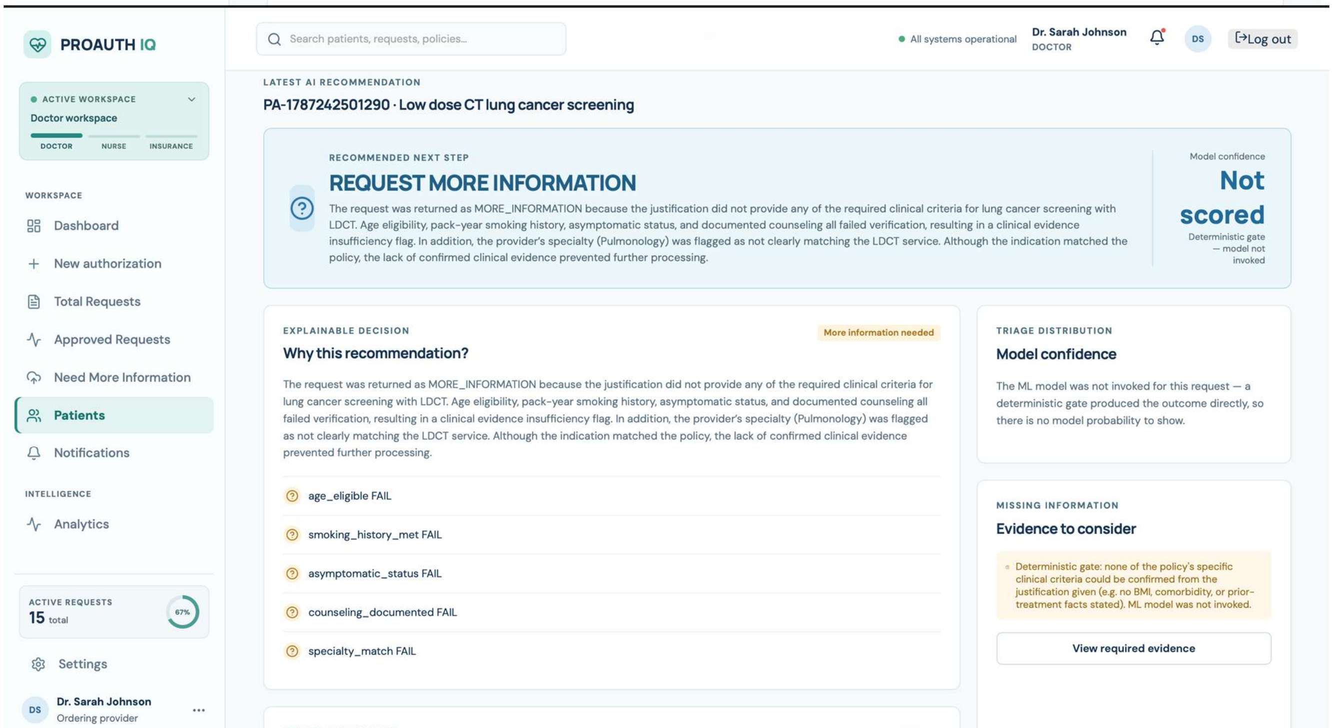
The proposed system combines **Policy RAG, Lang-Graph-based agentic AI, and machine learning**. When a doctor submits an authorization request, the system retrieves the relevant **Medicare coverage policy**, analyses the patient's clinical information and supporting documents through specialized agents, and converts the resulting evidence into structured features for the ML triage model.

The system produces three primary outcomes: **Approve, Pend for Nurse Review, or Request More Information**. It also provides explainable results by showing relevant policy criteria, supporting evidence, missing information, and prediction confidence.

The project uses **React and Vite** for the frontend, **Node.js and Express** for the backend, **PostgreSQL** for structured data, **Python and FastAPI** for AI/ML services, **LangGraph and RAG** for agent orchestration and policy retrieval, and **Groq LLM** for language-based reasoning. **Synthea** provides synthetic healthcare data for development and evaluation.

Appendix


PROAUTH IQ
All systems operational
Kenneth Young
USER / MEMBER
KY
Log out
ACTIVE WORKSPACE
Member portal
WORKSPACE
Profile
My Policy
Diagnosis
My Request
Decision
Documents
Timeline
MY REQUESTS
9 total
33%
Settings
Kenneth Young
Read-only access
DIAGNOSIS SPOTLIGHT
View only
Non-small cell lung cancer, stage IIIA, status post chemoradiation
Pulmonary System
C34.90
Affects the lungs and airways, involved in breathing and gas exchange.
Urgency
Routine
Clinical history
61-year-old male with stage IIIA non-small cell lung cancer status post concurrent chemoradiation, now 3 months out, presenting for restaging PET-CT to assess treatment response prior to determining further management.
RELATED AUTHORIZATION
Needs Review
PET-CT scan for oncologic restaging
REQUEST ID
PA-1787308365830
PROCEDURE CODE
78815
CARE SETTING
Outpatient
SUBMITTED
Aug 21, 4:02 PM
Decision: Pend for reviewer
The request was placed in pending review because no applicable coverage policy could be identified, triggering a deterministic gate that prevented any clinical criteria evaluation. Consequently, the ML model was not invoked. Required documents (Clinical notes and Prescription) were missing, and both indication_match and specialty_match criteria failed, as did diagnosis support. The request will await manual review.


PROAUTH IQ
Search patients, requests, policies...
All systems operational
Dr. Sarah Johnson
DOCTOR
DS
Log out
ACTIVE WORKSPACE
Doctor workspace
DOCTOR
NURSE
INSURANCE
WORKSPACE
Dashboard
New authorization
Total Requests
Approved Requests
Need More Information
Patients
Notifications
INTELLIGENCE
Analytics
ACTIVE REQUESTS
15 total
67%
Settings
Dr. Sarah Johnson
Ordering provider
LATEST AI RECOMMENDATION
PA-1787242501290 · Low dose CT lung cancer screening
RECOMMENDED NEXT STEP
REQUEST MORE INFORMATION
The request was returned as MORE_INFORMATION because the justification did not provide any of the required clinical criteria for lung cancer screening with LDCT. Age eligibility, pack-year smoking history, asymptomatic status, and documented counseling all failed verification, resulting in a clinical evidence insufficiency flag. In addition, the provider's specialty (Pulmonology) was flagged as not clearly matching the LDCT service. Although the indication matched the policy, the lack of confirmed clinical evidence prevented further processing.
Model confidence
Not scored
Deterministic gate — model not invoked
EXPLAINABLE DECISION
More information needed
Why this recommendation?
The request was returned as MORE_INFORMATION because the justification did not provide any of the required clinical criteria for lung cancer screening with LDCT. Age eligibility, pack-year smoking history, asymptomatic status, and documented counseling all failed verification, resulting in a clinical evidence insufficiency flag. In addition, the provider's specialty (Pulmonology) was flagged as not clearly matching the LDCT service. Although the indication matched the policy, the lack of confirmed clinical evidence prevented further processing.
age_eligible FAIL
smoking_history_met FAIL
asymptomatic_status FAIL
counseling_documented FAIL
specialty_match FAIL
POLICY COMPLIANCE
TRIAGE DISTRIBUTION
Model confidence
The ML model was not invoked for this request — a deterministic gate produced the outcome directly, so there is no model probability to show.
MISSING INFORMATION
Evidence to consider
Deterministic gate: none of the policy's specific clinical criteria could be confirmed from the justification given (e.g. no BMI, comorbidity, or prior-treatment facts stated). ML model was not invoked.
View required evidence

PROAUTH IQ

ACTIVE WORKSPACE
Doctor workspace
DOCTOR NURSE INSURANCE

WORKSPACE
- Dashboard
- New authorization
- Total Requests
- Approved Requests
- Need More Information
- Patients
- Notifications

INTELLIGENCE
- Analytics

ACTIVE REQUESTS
15 total 67%

Settings

DS Dr. Sarah Johnson
Ordering provider

Search patients, requests, policies...

All systems operational Dr. Sarah Johnson DOCTOR DS Log out

< Back to my requests

RESUBMIT FOR REVIEW

PA-1787242501290

Additional information was requested. Add what's needed below and resubmit — the AI pipeline re-evaluates automatically and can take up to two minutes.

Additional clinical justification

Add whatever detail addresses the reviewer's note — this is appended to your original justification, not a replacement for it.

Add supporting documents

Document	Accepted formats	Status
Clinical notes	PDF, DOCX, PNG, or JPG · Maximum file size 20 MB	Not attached
Lab reports	PDF, DOCX, PNG, or JPG · Maximum file size 20 MB	Not attached
Previous treatment records	PDF, DOCX, PNG, or JPG · Maximum file size 20 MB	Not attached
Prescriptions	PDF, DOCX, PNG, or JPG · Maximum file size 20 MB	Not attached
Imaging reports	PDF, DOCX, PNG, or JPG · Maximum file size 20 MB	Not attached

PROAUTH IQ

ACTIVE WORKSPACE
Doctor workspace
DOCTOR NURSE INSURANCE

WORKSPACE
- Dashboard
- New authorization
- Total Requests
- Approved Requests
- Need More Information
- Patients
- Notifications

INTELLIGENCE
- Analytics

ACTIVE REQUESTS
15 total 67%

Settings

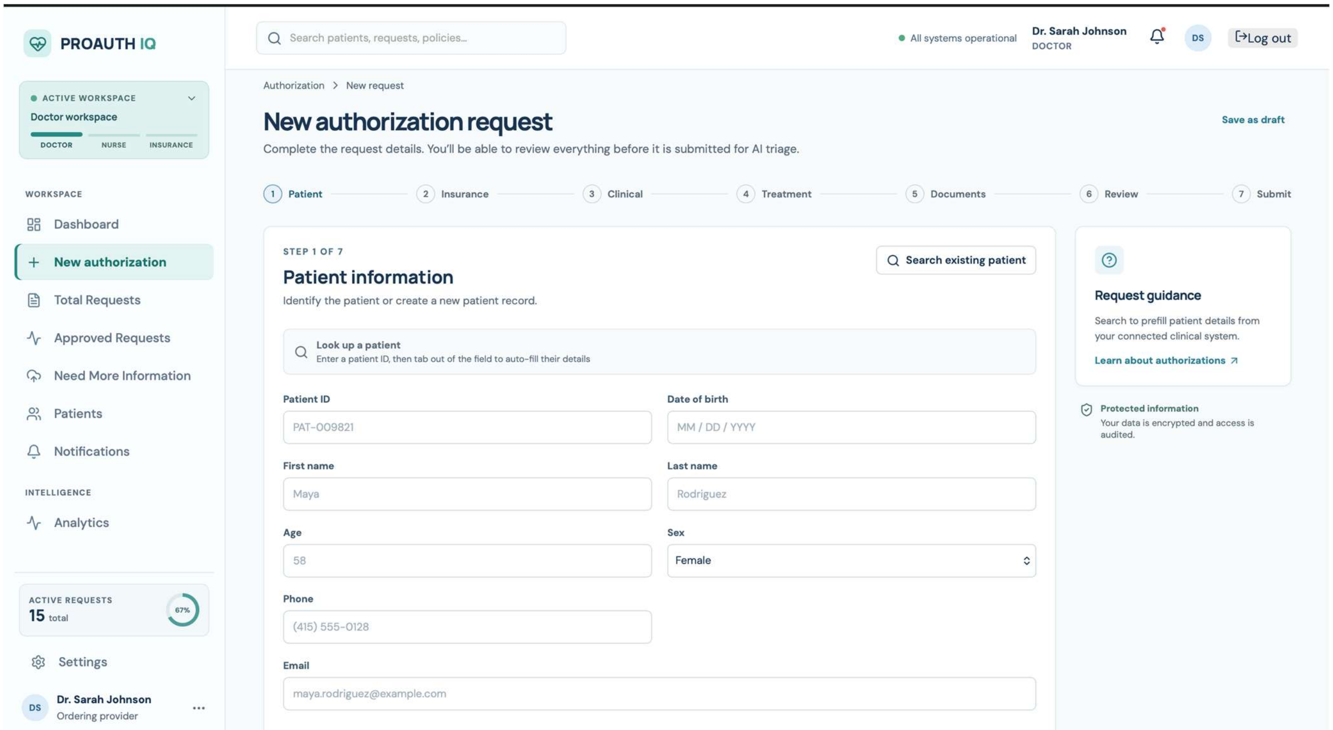
DS Dr. Sarah Johnson
Ordering provider

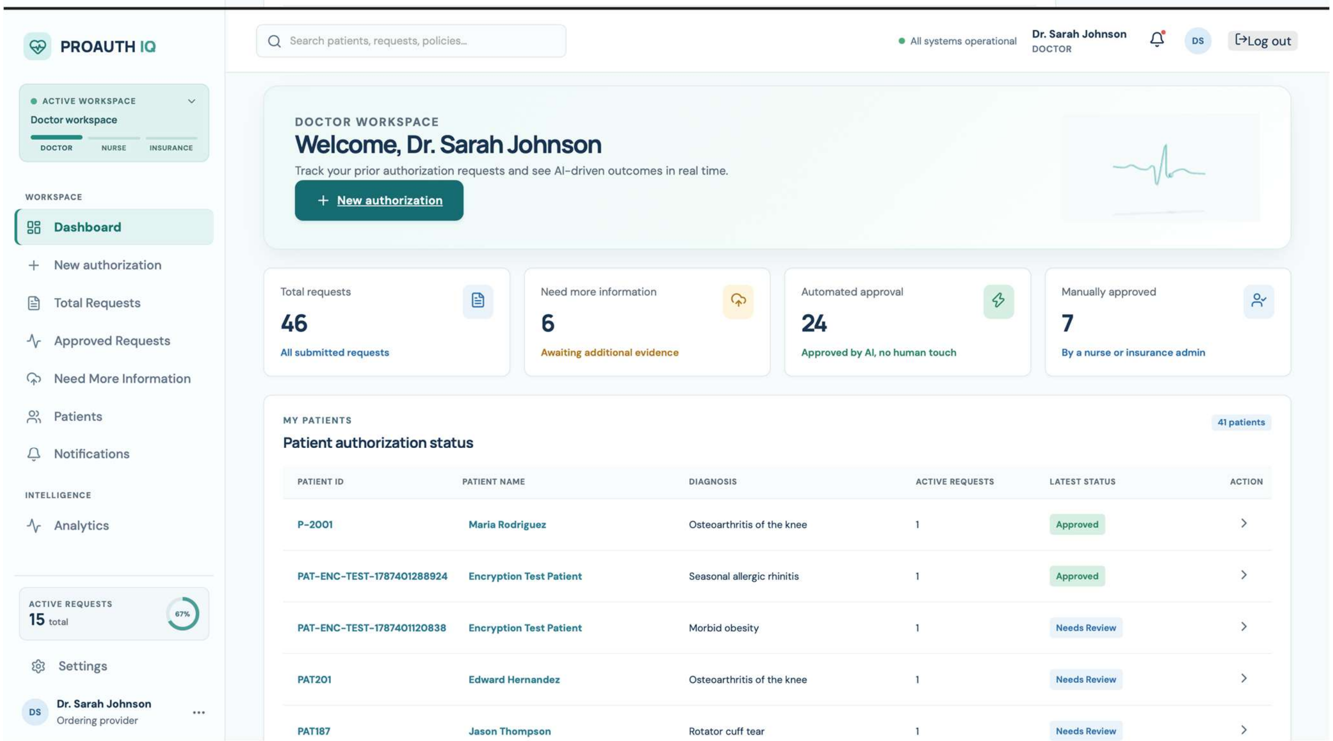
Search patients, requests, policies...

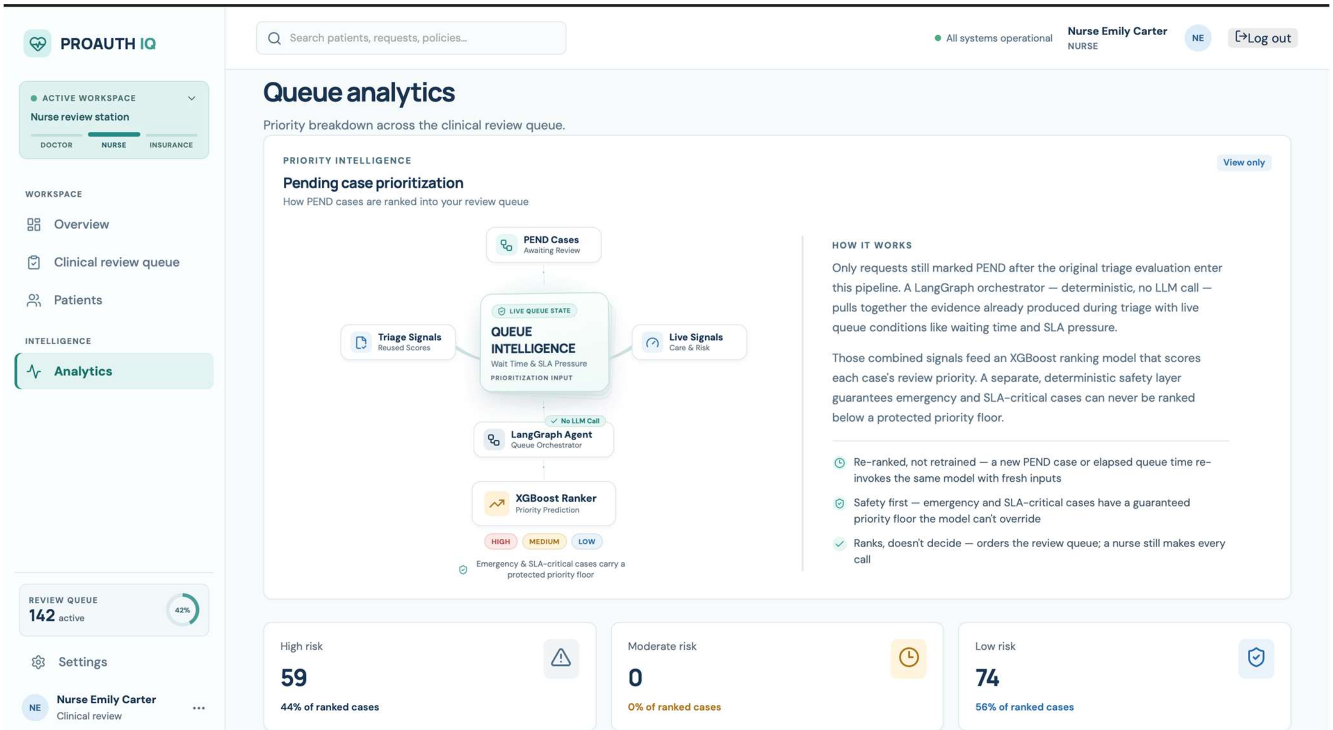
All systems operational Dr. Sarah Johnson DOCTOR DS Log out

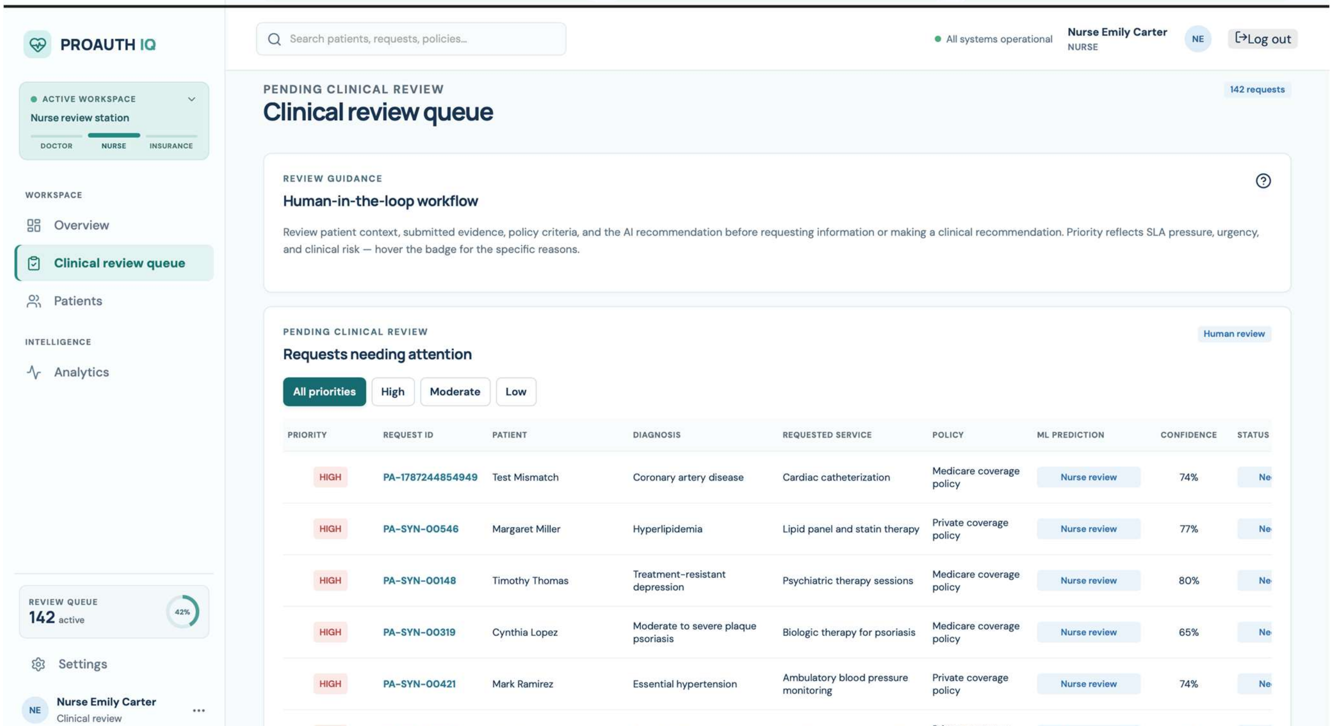
Need more information

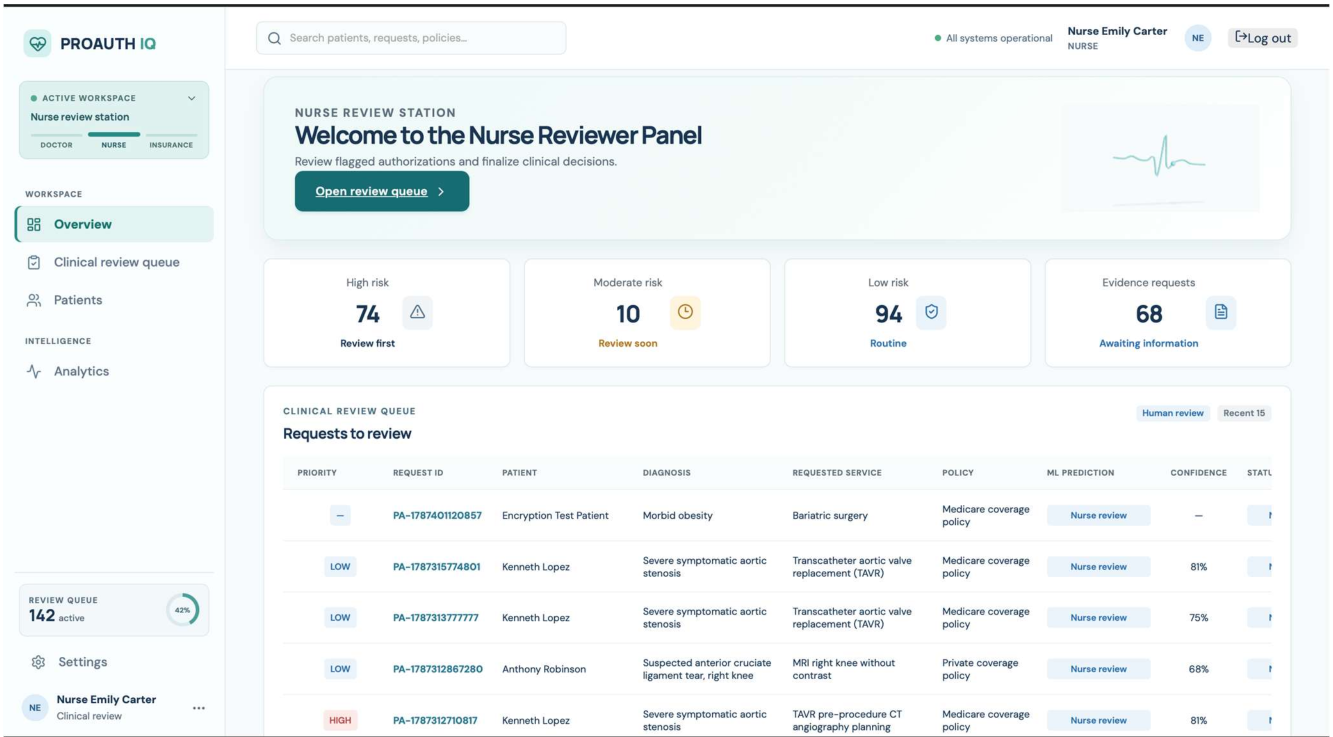
REQUEST ID	PATIENT	REQUESTED SERVICE	MISSING INFORMATION	MISSING DOCUMENTS	DATE REQUESTED	ACTION
PA-1787242501290	Linda Martinez	Low dose CT lung cancer sc...		Not specified	Aug 20, 9:45 PM	View · Upload info · Resubmit
PA-1787240785191	Unnamed patient	Specialist service		Not specified	Aug 20, 9:16 PM	View · Upload info · Resubmit
PA-1787238342175	Kenneth Young	Continuous glucose monitor		Not specified	Aug 20, 8:35 PM	View · Upload info · Resubmit
PA-1787236171261	Kenneth Young	MRI lower extremity		Not specified	Aug 20, 7:59 PM	View · Upload info · Resubmit
PA-1787235342749	Unnamed patient	Imaging		Not specified	Aug 20, 7:45 PM	View · Upload info · Resubmit
PA-SYN-00205	Ashley Martinez	Knee arthroscopy		Not specified	Aug 19, 11:12 AM	View · Upload info · Resubmit

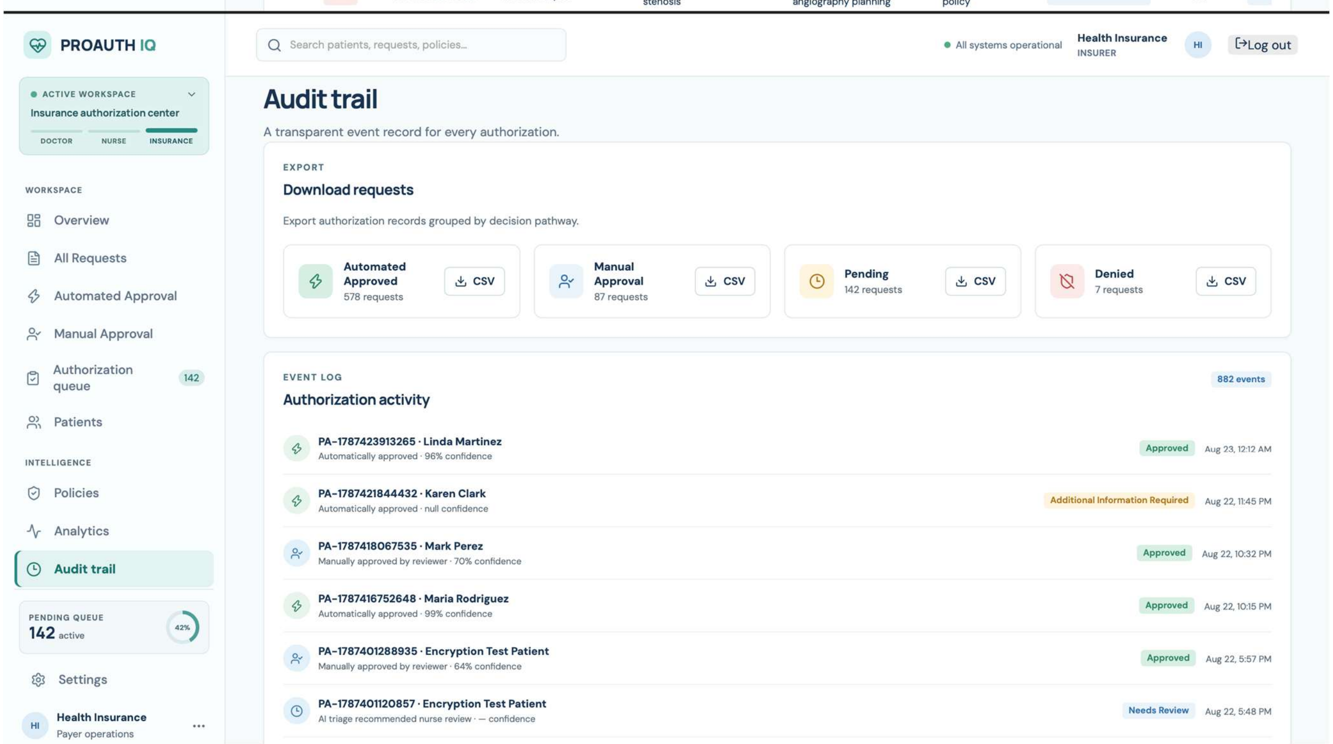

PROAUTH IQ
Search patients, requests, policies...
All systems operational
Dr. Sarah Johnson
DOCTOR
DS
Log out
ACTIVE WORKSPACE
Doctor workspace
DOCTOR
NURSE
INSURANCE
WORKSPACE
Dashboard
New authorization
Total Requests
Approved Requests
Need More Information
Patients
Notifications
INTELLIGENCE
Analytics
ACTIVE REQUESTS
15 total
67%
Settings
Dr. Sarah Johnson
Ordering provider
Authorization > New request
New authorization request
Save as draft
Complete the request details. You'll be able to review everything before it is submitted for AI triage.
1 Patient
2 Insurance
3 Clinical
4 Treatment
5 Documents
6 Review
7 Submit
STEP 1 OF 7
Patient information
Search existing patient
Identify the patient or create a new patient record.
Look up a patient
Enter a patient ID, then tab out of the field to auto-fill their details
Patient ID
PAT-009821
Date of birth
MM / DD / YYYY
First name
Maya
Last name
Rodriguez
Age
58
Sex
Female
Phone
(415) 555-0128
Email
maya.rodriguez@example.com
Request guidance
Search to prefill patient details from your connected clinical system.
Learn about authorizations
Protected information
Your data is encrypted and access is audited.

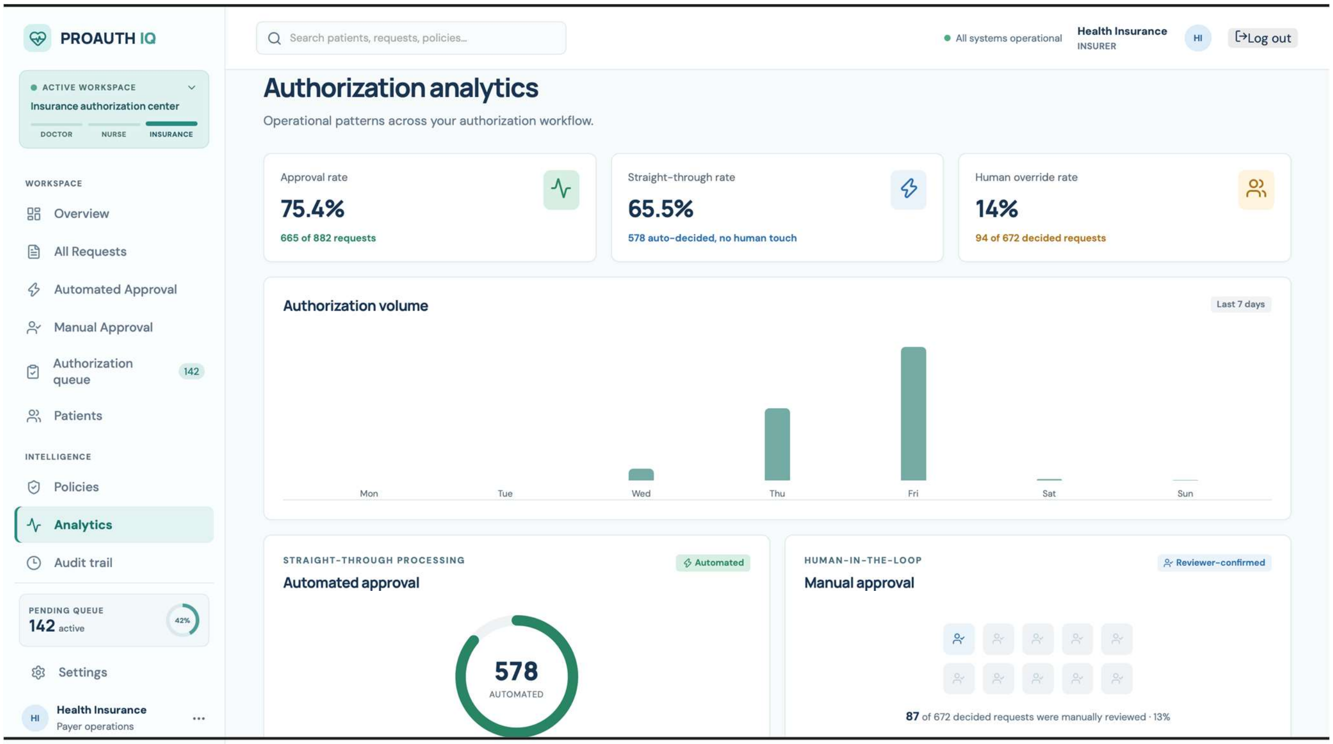

PROAUTH IQ
Search patients, requests, policies...
All systems operational
Dr. Sarah Johnson
DOCTOR
DS
Log out
ACTIVE WORKSPACE
Doctor workspace
DOCTOR
NURSE
INSURANCE
WORKSPACE
Dashboard
New authorization
Total Requests
Approved Requests
Need More Information
Patients
Notifications
INTELLIGENCE
Analytics
ACTIVE REQUESTS
15 total
67%
Settings
Dr. Sarah Johnson
Ordering provider
DOCTOR WORKSPACE
Welcome, Dr. Sarah Johnson
Track your prior authorization requests and see AI-driven outcomes in real time.
New authorization
Total requests
46
All submitted requests
Need more information
6
Awaiting additional evidence
Automated approval
24
Approved by AI, no human touch
Manually approved
7
By a nurse or insurance admin
MY PATIENTS
41 patients
Patient authorization status
PATIENT ID
PATIENT NAME
DIAGNOSIS
ACTIVE REQUESTS
LATEST STATUS
ACTION
P-2001
Maria Rodriguez
Osteoarthritis of the knee
1
Approved
PAT-ENC-TEST-1787401288924
Encryption Test Patient
Seasonal allergic rhinitis
1
Approved
PAT-ENC-TEST-1787401120838
Encryption Test Patient
Morbid obesity
1
Needs Review
PAT201
Edward Hernandez
Osteoarthritis of the knee
1
Needs Review
PAT187
Jason Thompson
Rotator cuff tear
1
Needs Review

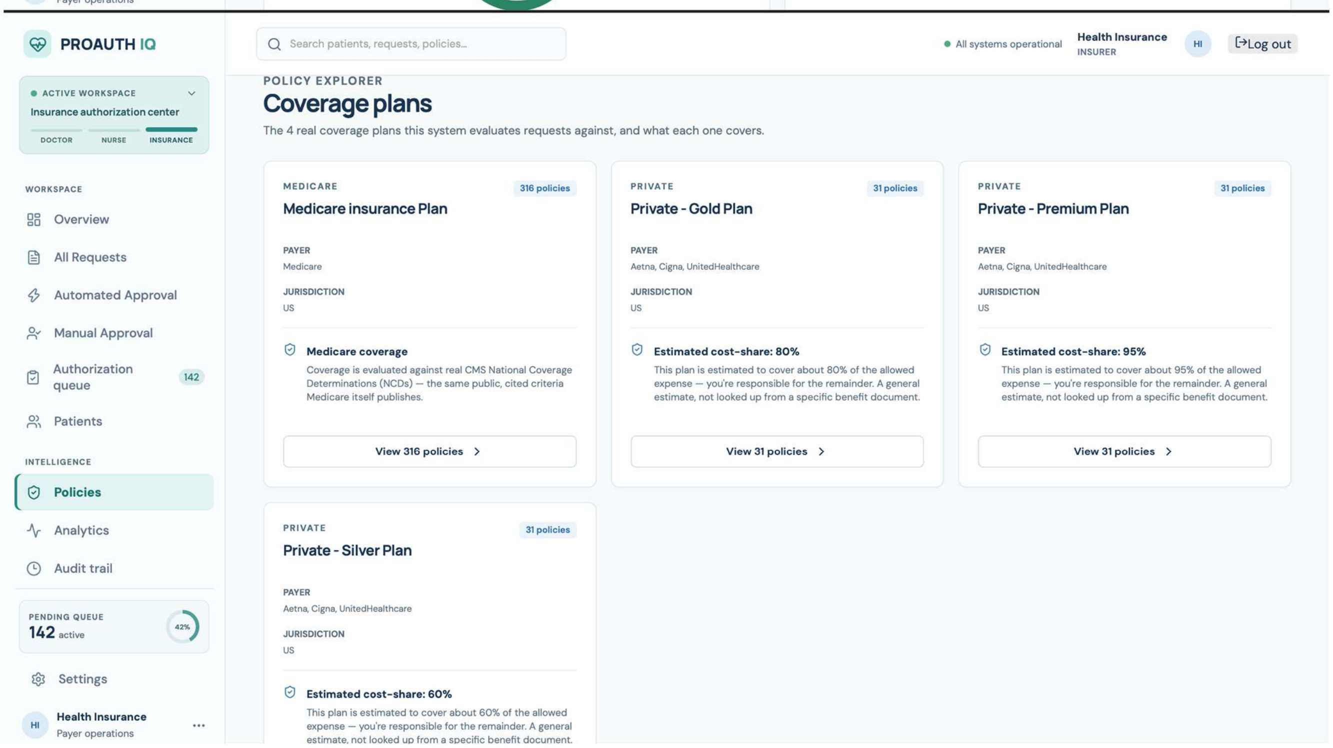

PROAUTH IQ
Search patients, requests, policies...
All systems operational
Nurse Emily Carter
NURSE
NE
Log out
ACTIVE WORKSPACE
Nurse review station
DOCTOR
NURSE
INSURANCE
WORKSPACE
Overview
Clinical review queue
Patients
INTELLIGENCE
Analytics
REVIEW QUEUE
142 active
42%
Settings
Nurse Emily Carter
Clinical review
Queue analytics
Priority breakdown across the clinical review queue.
PRIORITY INTELLIGENCE
Pending case prioritization
How PEND cases are ranked into your review queue
View only
PEND Cases
Awaiting Review
LIVE QUEUE STATE
QUEUE INTELLIGENCE
Wait Time & SLA Pressure
PRIORITIZATION INPUT
Triage Signals
Reused Scores
Live Signals
Care & Risk
No LLM Call
LangGraph Agent
Queue Orchestrator
XGBoost Ranker
Priority Prediction
HIGH
MEDIUM
LOW
Emergency & SLA-critical cases carry a protected priority floor
HOW IT WORKS
Only requests still marked PEND after the original triage evaluation enter this pipeline. A LangGraph orchestrator — deterministic, no LLM call — pulls together the evidence already produced during triage with live queue conditions like waiting time and SLA pressure.
Those combined signals feed an XGBoost ranking model that scores each case's review priority. A separate, deterministic safety layer guarantees emergency and SLA-critical cases can never be ranked below a protected priority floor.
Re-ranked, not retrained — a new PEND case or elapsed queue time re-invokes the same model with fresh inputs
Safety first — emergency and SLA-critical cases have a guaranteed priority floor the model can't override
Ranks, doesn't decide — orders the review queue; a nurse still makes every call
High risk
59
44% of ranked cases
Moderate risk
0
0% of ranked cases
Low risk
74
56% of ranked cases

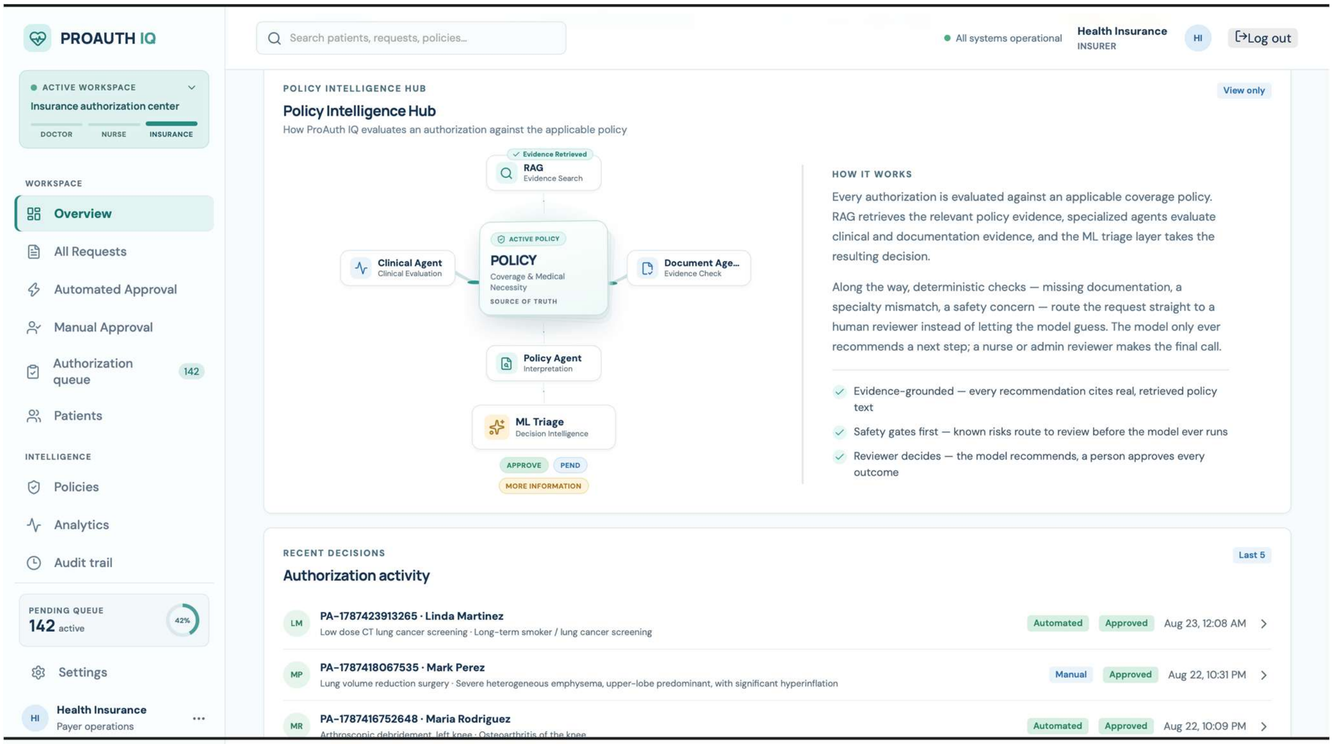

PROAUTH IQ
Search patients, requests, policies...
All systems operational
Nurse Emily Carter
NURSE
NE
Log out
ACTIVE WORKSPACE
Nurse review station
DOCTOR
NURSE
INSURANCE
WORKSPACE
Overview
Clinical review queue
Patients
INTELLIGENCE
Analytics
REVIEW QUEUE
142 active
42%
Settings
Nurse Emily Carter
Clinical review
PENDING CLINICAL REVIEW
Clinical review queue
142 requests
REVIEW GUIDANCE
Human-in-the-loop workflow
Review patient context, submitted evidence, policy criteria, and the AI recommendation before requesting information or making a clinical recommendation. Priority reflects SLA pressure, urgency, and clinical risk — hover the badge for the specific reasons.
PENDING CLINICAL REVIEW
Requests needing attention
Human review
All priorities
High
Moderate
Low
PRIORITY | REQUEST ID | PATIENT | DIAGNOSIS | REQUESTED SERVICE | POLICY | ML PREDICTION | CONFIDENCE | STATUS
HIGH | PA-1787244854949 | Test Mismatch | Coronary artery disease | Cardiac catheterization | Medicare coverage policy | Nurse review | 74% | Ne
HIGH | PA-SYN-00546 | Margaret Miller | Hyperlipidemia | Lipid panel and statin therapy | Private coverage policy | Nurse review | 77% | Ne
HIGH | PA-SYN-00148 | Timothy Thomas | Treatment-resistant depression | Psychiatric therapy sessions | Medicare coverage policy | Nurse review | 80% | Ne
HIGH | PA-SYN-00319 | Cynthia Lopez | Moderate to severe plaque psoriasis | Biologic therapy for psoriasis | Medicare coverage policy | Nurse review | 65% | Ne
HIGH | PA-SYN-00421 | Mark Ramirez | Essential hypertension | Ambulatory blood pressure monitoring | Private coverage policy | Nurse review | 74% | Ne
HIGH | PA-SYN-00384 | Gary Ramirez | Type 2 diabetes mellitus | Continuous glucose monitor | Private coverage | Nurse review | 77% | Ne

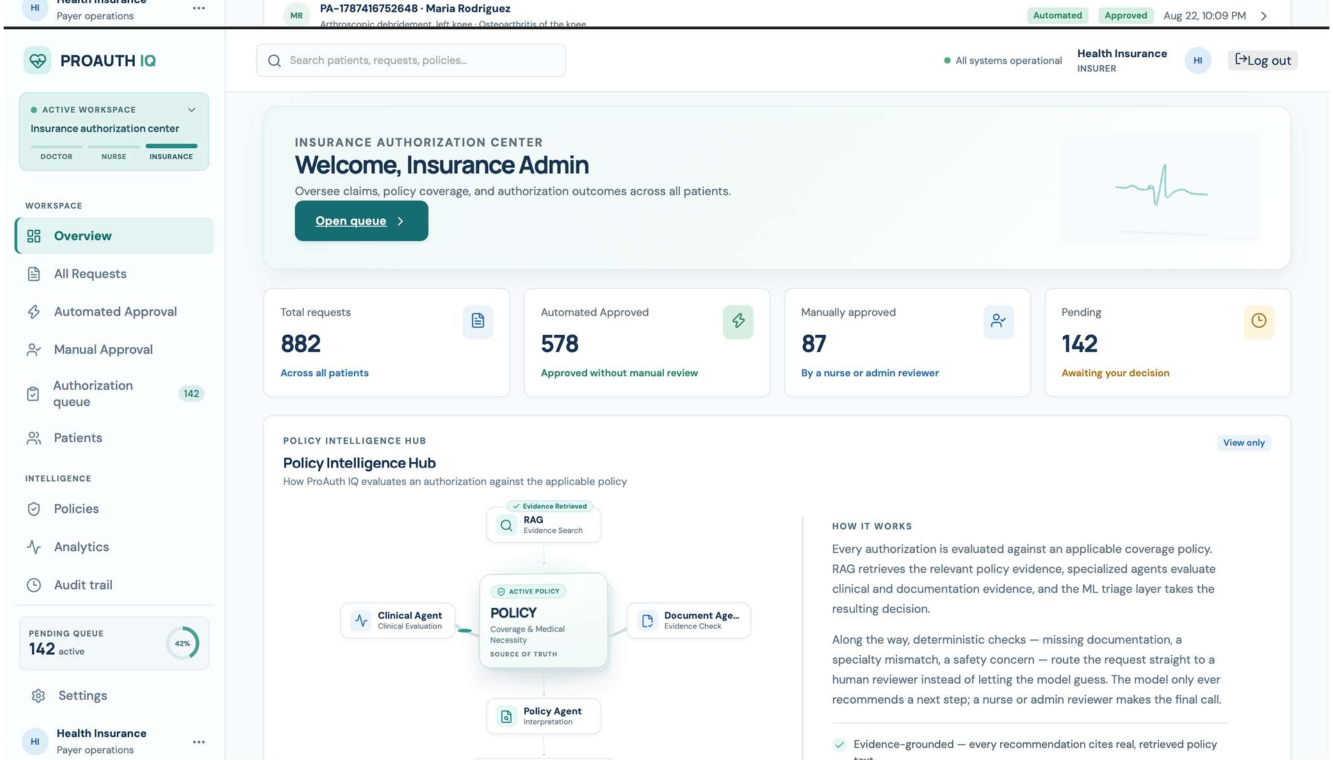

PROAUTH IQ
Search patients, requests, policies...
All systems operational
Nurse Emily Carter
NURSE
NE
Log out
ACTIVE WORKSPACE
Nurse review station
DOCTOR
NURSE
INSURANCE
WORKSPACE
Overview
Clinical review queue
Patients
INTELLIGENCE
Analytics
REVIEW QUEUE
142 active
42%
Settings
Nurse Emily Carter
Clinical review
NURSE REVIEW STATION
Welcome to the Nurse Reviewer Panel
Review flagged authorizations and finalize clinical decisions.
Open review queue
High risk
74
Review first
Moderate risk
10
Review soon
Low risk
94
Routine
Evidence requests
68
Awaiting information
CLINICAL REVIEW QUEUE
Requests to review
Human review
Recent 15
PRIORITY
REQUEST ID
PATIENT
DIAGNOSIS
REQUESTED SERVICE
POLICY
ML PREDICTION
CONFIDENCE
STATU
—
PA-1787401120857
Encryption Test Patient
Morbid obesity
Bariatric surgery
Medicare coverage policy
Nurse review
—
LOW
PA-1787315774801
Kenneth Lopez
Severe symptomatic aortic stenosis
Transcatheter aortic valve replacement (TAVR)
Medicare coverage policy
Nurse review
81%
LOW
PA-1787313777777
Kenneth Lopez
Severe symptomatic aortic stenosis
Transcatheter aortic valve replacement (TAVR)
Medicare coverage policy
Nurse review
75%
LOW
PA-1787312867280
Anthony Robinson
Suspected anterior cruciate ligament tear, right knee
MRI right knee without contrast
Private coverage policy
Nurse review
68%
HIGH
PA-1787312710817
Kenneth Lopez
Severe symptomatic aortic stenosis
TAVR pre-procedure CT angiography planning
Medicare coverage policy
Nurse review
81%

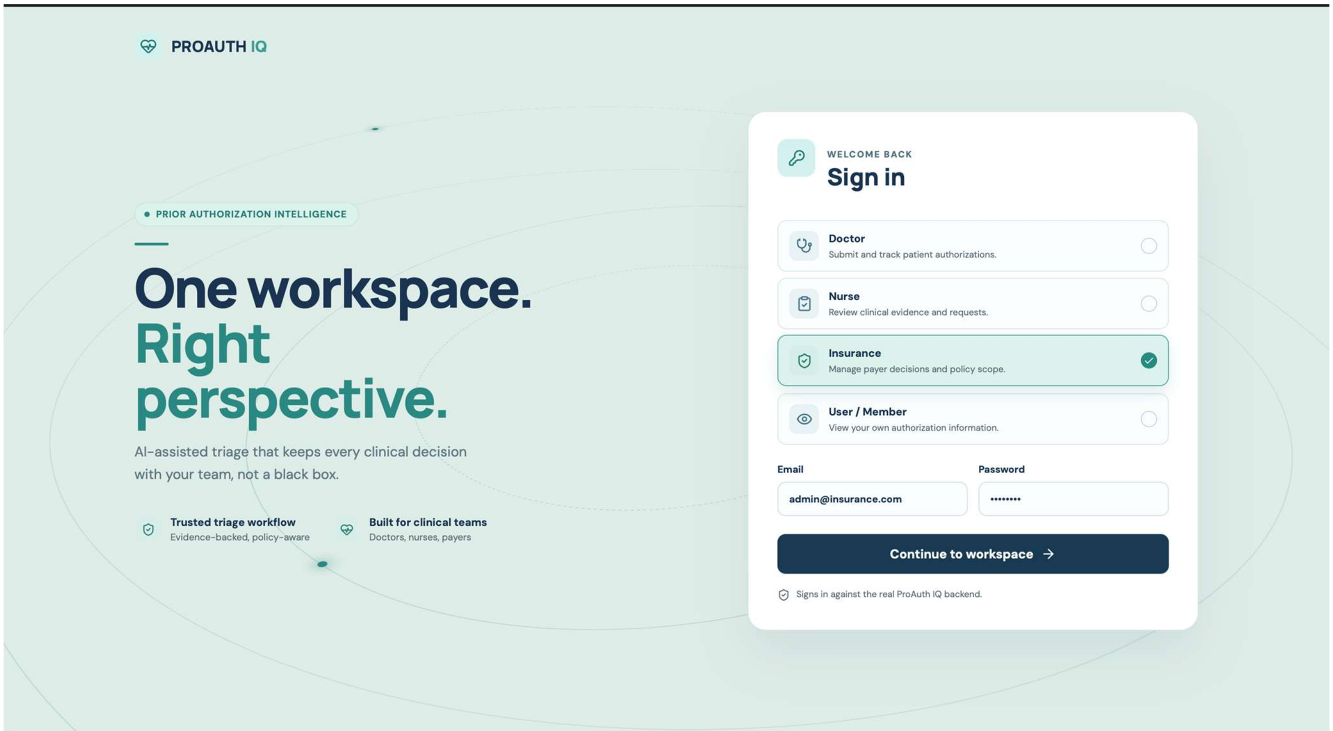

PROAUTH IQ
Search patients, requests, policies...
All systems operational
Health Insurance
INSURER
HI
Log out
ACTIVE WORKSPACE
Insurance authorization center
DOCTOR
NURSE
INSURANCE
WORKSPACE
Overview
All Requests
Automated Approval
Manual Approval
Authorization queue
142
Patients
INTELLIGENCE
Policies
Analytics
Audit trail
PENDING QUEUE
142 active
42%
Settings
Health Insurance
Payer operations
Audit trail
A transparent event record for every authorization.
EXPORT
Download requests
Export authorization records grouped by decision pathway.
Automated Approved
578 requests
CSV
Manual Approval
87 requests
CSV
Pending
142 requests
CSV
Denied
7 requests
CSV
EVENT LOG
Authorization activity
882 events
PA-1787423913265 · Linda Martinez
Automatically approved · 96% confidence
Approved
Aug 23, 12:12 AM
PA-1787421844432 · Karen Clark
Automatically approved · null confidence
Additional Information Required
Aug 22, 11:45 PM
PA-1787418067535 · Mark Perez
Manually approved by reviewer · 70% confidence
Approved
Aug 22, 10:32 PM
PA-1787416752648 · Maria Rodriguez
Automatically approved · 99% confidence
Approved
Aug 22, 10:15 PM
PA-1787401288935 · Encryption Test Patient
Manually approved by reviewer · 64% confidence
Approved
Aug 22, 5:57 PM
PA-1787401120857 · Encryption Test Patient
AI triage recommended nurse review · — confidence
Needs Review
Aug 22, 5:48 PM


PROAUTH IQ
ACTIVE WORKSPACE
Insurance authorization center
DOCTOR
NURSE
INSURANCE
WORKSPACE
Overview
All Requests
Automated Approval
Manual Approval
Authorization queue
142
Patients
INTELLIGENCE
Policies
Analytics
Audit trail
PENDING QUEUE
142 active
42%
Settings
Health Insurance
Payer operations
HI
Search patients, requests, policies...
All systems operational
Health Insurance
INSURER
Log out
Authorization analytics
Operational patterns across your authorization workflow.
Approval rate
75.4%
665 of 882 requests
Straight-through rate
65.5%
578 auto-decided, no human touch
Human override rate
14%
94 of 672 decided requests
Authorization volume
Last 7 days
Mon
Tue
Wed
Thu
Fri
Sat
Sun
STRAIGHT-THROUGH PROCESSING
Automated
Automated approval
578
AUTOMATED
HUMAN-IN-THE-LOOP
Reviewer-confirmed
Manual approval
87 of 672 decided requests were manually reviewed · 13%


PROAUTH IQ
ACTIVE WORKSPACE
Insurance authorization center
DOCTOR
NURSE
INSURANCE
WORKSPACE
Overview
All Requests
Automated Approval
Manual Approval
Authorization queue
142
Patients
INTELLIGENCE
Policies
Analytics
Audit trail
PENDING QUEUE
142 active
42%
Settings
Health Insurance
Payer operations
HI
Search patients, requests, policies...
All systems operational
Health Insurance
INSURER
Log out
POLICY EXPLORER
Coverage plans
The 4 real coverage plans this system evaluates requests against, and what each one covers.
MEDICARE
316 policies
Medicare insurance Plan
PAYER
Medicare
JURISDICTION
US
Medicare coverage
Coverage is evaluated against real CMS National Coverage Determinations (NCDs) — the same public, cited criteria Medicare itself publishes.
View 316 policies
PRIVATE
31 policies
Private - Gold Plan
PAYER
Aetna, Cigna, UnitedHealthcare
JURISDICTION
US
Estimated cost-share: 80%
This plan is estimated to cover about 80% of the allowed expense — you're responsible for the remainder. A general estimate, not looked up from a specific benefit document.
View 31 policies
PRIVATE
31 policies
Private - Premium Plan
PAYER
Aetna, Cigna, UnitedHealthcare
JURISDICTION
US
Estimated cost-share: 95%
This plan is estimated to cover about 95% of the allowed expense — you're responsible for the remainder. A general estimate, not looked up from a specific benefit document.
View 31 policies
PRIVATE
31 policies
Private - Silver Plan
PAYER
Aetna, Cigna, UnitedHealthcare
JURISDICTION
US
Estimated cost-share: 60%
This plan is estimated to cover about 60% of the allowed expense — you're responsible for the remainder. A general estimate, not looked up from a specific benefit document.


PROAUTH IQ
Search patients, requests, policies...
All systems operational
Health Insurance
INSURER
HI
Log out
ACTIVE WORKSPACE
Insurance authorization center
DOCTOR NURSE INSURANCE
WORKSPACE
Overview
All Requests
Automated Approval
Manual Approval
Authorization queue 142
Patients
INTELLIGENCE
Policies
Analytics
Audit trail
PENDING QUEUE
142 active
42%
Settings
Health Insurance
Payer operations
POLICY INTELLIGENCE HUB
Policy Intelligence Hub
How ProAuth IQ evaluates an authorization against the applicable policy
View only
Evidence Retrieved
RAG
Evidence Search
ACTIVE POLICY
POLICY
Coverage & Medical Necessity
SOURCE OF TRUTH
Clinical Agent
Clinical Evaluation
Document Age...
Evidence Check
Policy Agent
Interpretation
ML Triage
Decision Intelligence
APPROVE
PEND
MORE INFORMATION
HOW IT WORKS
Every authorization is evaluated against an applicable coverage policy. RAG retrieves the relevant policy evidence, specialized agents evaluate clinical and documentation evidence, and the ML triage layer takes the resulting decision.
Along the way, deterministic checks — missing documentation, a specialty mismatch, a safety concern — route the request straight to a human reviewer instead of letting the model guess. The model only ever recommends a next step; a nurse or admin reviewer makes the final call.
Evidence-grounded — every recommendation cites real, retrieved policy text
Safety gates first — known risks route to review before the model ever runs
Reviewer decides — the model recommends, a person approves every outcome
RECENT DECISIONS
Authorization activity
Last 5
LM PA-1787423913265 · Linda Martinez
Low dose CT lung cancer screening · Long-term smoker / lung cancer screening
Automated Approved Aug 23, 12:08 AM
MP PA-1787418067535 · Mark Perez
Lung volume reduction surgery · Severe heterogeneous emphysema, upper-lobe predominant, with significant hyperinflation
Manual Approved Aug 22, 10:31 PM
MR PA-1787416752648 · Maria Rodriguez
Automated Approved Aug 22, 10:09 PM


PROAUTH IQ
Search patients, requests, policies...
All systems operational
Health Insurance
INSURER
HI
Log out
ACTIVE WORKSPACE
Insurance authorization center
DOCTOR NURSE INSURANCE
WORKSPACE
Overview
All Requests
Automated Approval
Manual Approval
Authorization queue 142
Patients
INTELLIGENCE
Policies
Analytics
Audit trail
PENDING QUEUE
142 active
42%
Settings
Health Insurance
Payer operations
INSURANCE AUTHORIZATION CENTER
Welcome, Insurance Admin
Oversee claims, policy coverage, and authorization outcomes across all patients.
Open queue
Total requests
882
Across all patients
Automated Approved
578
Approved without manual review
Manually approved
87
By a nurse or admin reviewer
Pending
142
Awaiting your decision
POLICY INTELLIGENCE HUB
Policy Intelligence Hub
How ProAuth IQ evaluates an authorization against the applicable policy
View only
Evidence Retrieved
RAG
Evidence Search
ACTIVE POLICY
POLICY
Coverage & Medical Necessity
SOURCE OF TRUTH
Clinical Agent
Clinical Evaluation
Document Age...
Evidence Check
Policy Agent
Interpretation
HOW IT WORKS
Every authorization is evaluated against an applicable coverage policy. RAG retrieves the relevant policy evidence, specialized agents evaluate clinical and documentation evidence, and the ML triage layer takes the resulting decision.
Along the way, deterministic checks — missing documentation, a specialty mismatch, a safety concern — route the request straight to a human reviewer instead of letting the model guess. The model only ever recommends a next step; a nurse or admin reviewer makes the final call.
Evidence-grounded — every recommendation cites real, retrieved policy


PROAUTH IQ
PRIOR AUTHORIZATION INTELLIGENCE
One workspace.
Right perspective.
AI-assisted triage that keeps every clinical decision with your team, not a black box.
Trusted triage workflow
Evidence-backed, policy-aware
Built for clinical teams
Doctors, nurses, payers
WELCOME BACK
Sign in
Doctor
Submit and track patient authorizations.
Nurse
Review clinical evidence and requests.
Insurance
Manage payer decisions and policy scope.
User / Member
View your own authorization information.
Email
admin@insurance.com
Password
Continue to workspace
Signs in against the real ProAuth IQ backend.